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CORRELATION AND GEOCHRONOLOGY OF MIDDLE EOCENE STRATA FROM THE WESTERN UNITED STATES

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ABSTRACT

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In this study I integrate marine and continental biostratigraphy, magnetic polarity stratigraphy, and radioisotopic chronology in a synthetic correlation of middle Eocene strata from the western United States. More than 2000 m of section were sampled from volcaniclastic deposits, Aycross and Tepee Trail Formations (northwestern Wyoming); lacustrine and fluviatile deposits, Washakie Formation (southwestern Wyoming); and intertonguing marine and continental strata, La Jolla and Poway Groups (San Diego area, California).

Detailed demagnetization studies on numerous pilot samples from all three field areas revealed moderately complex magnetizations. Most sample NRM's are dominated by a strong normal polarity magnetic component; alternating field demagnetization does not consistently isolate the primary magnetization. High blocking temperature hematite is frequently a significant carrier of remanence. Therefore, most samples were subjected to detailed, stepwise alternating field *and* thermal demagnetization to 600—650°C.

The East Fork Basin area (northwestern Wyoming) magnetic polarity sequence consists of five major polarity intervals, A— to E—; the thick Washakie Formation sequence contains four, A+ to D—; and the San Diego area sequence has four polarity intervals, A— to D+.

A new biochronologic interval, the Shoshonian Land Mammal Subage (Earliest Uintan), is defined and characterized. In all three field areas Shoshonian (Earliest Uintan) faunas and the Bridgerian/Uintan boundary occur within a single long reversed polarity interval. Correlation of marine biostratigraphy between the San Diego area section and deep sea sections allows precise identification of San Diego polarity interval B+ as Chron C21N. Therefore, the Bridgerian/Uintan boundary and earliest Uintan faunas occur within the reversed interval of Chron C20R. High-temperature, K—Ar dates bracketing this horizon in northwestern Wyoming provide an age estimate of approximately 49.5 Ma for the top of Chron C21N and 49 Ma for the Bridgerian/Uintan boundary.

Berggren et al. (1985) use the age estimate of 49.5 Ma for the top of anomaly 21 (younger boundary of Chron C21N) as one calibration point for the generation of a Paleogene geochronology. The methodology and conclusions of this geochronology are compared to those of other recent geochronologies. Data from independent studies integrating high temperature radioisotopic dates, biochronology, and magnetochronology are used to test the validity of the Berggren et al. (1985) geochronology.

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INTRODUCTION AND METHODOLOGY

Temporal correlation of geographically and environmentally distinct rock units is an important goal of stratigraphic research. It has previously been difficult to assign terrestrial strata to equivalent time units of less than Epoch magnitude. Investigation of the biostratigraphy, magnetostratigraphy and radioisotopic chronology of Eocene sequences in Wyoming and California may greatly increase the temporal precision of intracontinental correlations.

A multidisciplinary approach is extremely useful in solving problems in paleontology, geology, and other aspects of earth history. A broad perspective and diverse sources of data often contribute information and insights essential to the resolution of scientific inquiries. In this study I attempt to integrate marine and continental biostratigraphy, magnetic polarity stratigraphy, and radioisotopic chronology in a synthetic correlation of sedimentary sequences from the western United States. A few detailed, multidisciplinary correlation studies have been attempted for late Tertiary deposits, but similar projects have rarely been undertaken for the early Tertiary.

This study emphasizes original paleomagnetic analyses of strata containing approximately equivalent mammalian faunas, and at least one other source of temporal data. The volcaniclastic deposits of the presently high-altitude Aycross and Tepee Trail Formations of northwestern Wyoming include a large mammal fauna, radioisotopic dates and paleobotanical information. Faunas from the extensive lacustrine and fluviatile deposits of the Washakie Formation of southwestern Wyoming have been collected and published upon by paleontologists since the late 19th century. Paleobotanical collections are also available from the Washakie Formation. Coastal, continental deposits in San Diego, California contain previously described mammalian faunas and a pollen flora, and intertongue with fossiliferous marine strata.

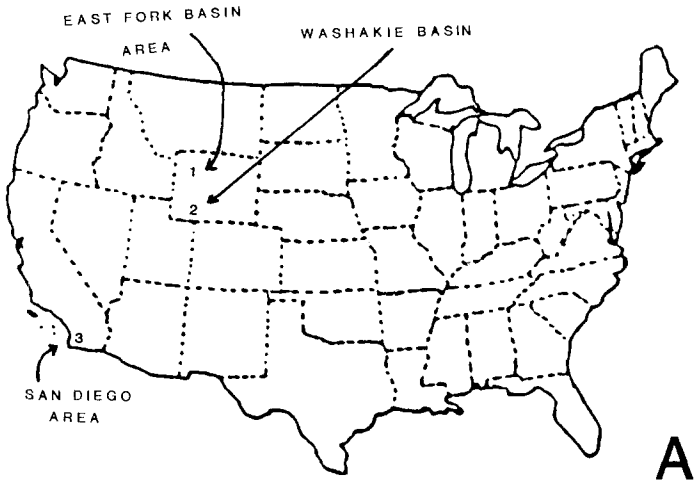
These three field areas (Fig.1) were selected to maximize the potential data sources in each area. The lithologies, environments of deposition, geographical locations, paleoclimatic conditions, etc. are intentionally widely varied so as to minimize the effects of local irregularities and biases and to broaden the scope and applicability of the correlations. A successful analysis of this type allows a more precise, intracontinental correlation of *widely* separated geographic localities than has ever been possible.

In this study I evaluate whether it is possible to:

(1) increase the resolution of temporal correlations for terrestrial sequences by using many independent data sources;

(2) successfully interface the marine macro-invertebrate, and planktonic and benthonic microfossil, chronologies with the vertebrate-derived continental time scale;

(3) apply continentally-derived, high temperature radioisotopic ages to dating Eocene marine magnetic anomalies (and anomaly chrons), via magnetostratigraphy and intertonguing marine and continental deposits.



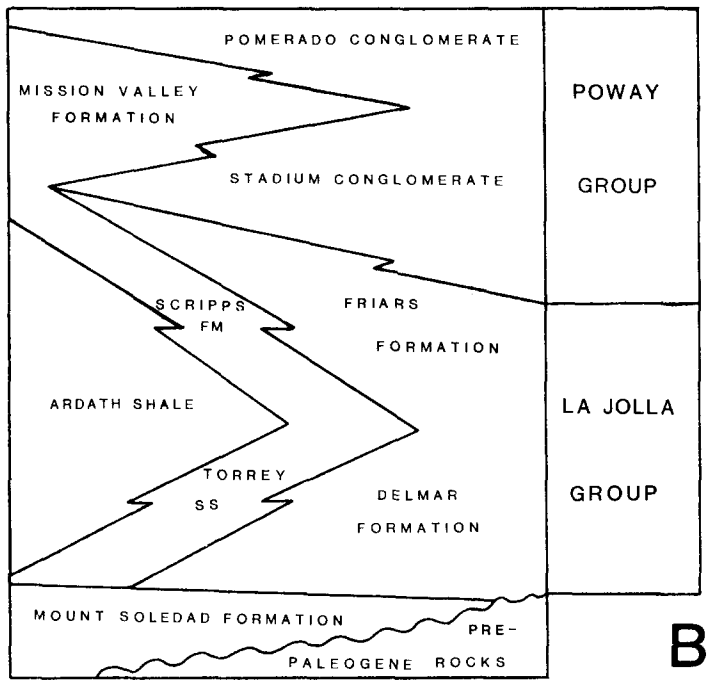


Fig.1. A. Locality map for the East Fork Basin area, Wyoming; Washakie Basin, Wyoming; and San Diego area, California study areas. B. Diagramatic representation of the stratigraphic relationships of the San Diego area sequence (modified after Kennedy, 1975).

(4) apply the resulting age estimates for Eocene polarity chrons to develop a more precise Paleogene geochronology (Berggren et al., 1985). Information from several recent studies that integrate paleomagnetic, biochronologic, and high temperature radioisotopic data is used to test the validity of the Berggren et al. (1985) geochronology.

Traditionally, radioisotopic chronology has been used to locate the approximate temporal position of a terrestrial magnetostratigraphy pattern within the standard geomagnetic polarity time scale (e.g. Johnson et al., 1975; MacFadden, 1977). In this study, however, the multiple data sources permit the use of biostratigraphy to determine the correlation of terrestrial magnetostratigraphies to a standard geomagnetic polarity time scale (GMPTS). This then leaves the radioisotopic chronology data available for independent (and non-circular) dating of the magnetic polarity and biostratigraphic boundaries within these sections.

I construct my "correlation web" (Berry, 1968) by (1) determining the magnetic polarity sequences in each section, (2) using mammalian biostratigraphy to correlate intracontinentally every sequence to the others, and to determine which polarity intervals are equivalent in the sequences from California, northwestern Wyoming, and southwestern Wyoming, (3) using the direct marine biostratigraphic bracketing (in southern California) of the same mammalian biostratigraphic zones and boundaries used in the intracontinental correlation, to interface marine and continental bio- and magnetostratigraphy, (4) using the known correlation between marine biostratigraphy and magnetostratigraphy (from DSDP studies, and from work on pelagic sequences in Italy) to identify uniquely the magnetic polarity intervals in these sections (precisely correlate the polarity intervals in my study sequences with magnetic polarity chrons in the standard GMPTS), and (5) determining ages of the biochronologic and polarity chron boundaries, independently of the correlation to the GMPTS, by using terrestrial radioisotopic data bracketing these boundaries. Development of this correlation network admittedly involves a complex sequence of steps, each of which includes its own set of errors, methodological assumptions and constraints. Errors must be minimized and assumptions must be rigorously evaluated, but it is the careful integration of data from a series of independent sources that is also the strength of such a synthetic correlation network. Only in this manner can the most complex geological problems be solved with increasing accuracy and refinement.

Several terms and abbreviations used within the body of this paper are defined here. In the reporting of radioisotopic dates, all dates have been calculated, or recalculated, using the new IUGS decay and abundance constants published in Steiger and Jager (1977).

Where stratigraphic or distance measurements were reported in non-metric (feet or miles) units in the original literature I retain these measurements in the text. Metric corrections are given in parentheses following these measurements.

In the paleomagnetics and geochronology discussions "geomagnetic polarity time scale" is abbreviated as GMPTS, and the abbreviations "y" or "o" refer to the younger or older boundaries, respectively, of magnetic polarity intervals. The magnetic polarity chron nomenclature of LaBrecque et al. (1983) is used as modified in Berggren et al. (1985); the prefix C indicates the Chron associated with a particular numbered magnetic anomaly, the suffix N (e.g. Chron C21N) refers to the dominantly normal polarity chron interval(s) associated with the numbered magnetic anomaly (e.g. Anomaly 21), and the suffix R (e.g. Chron C21R) refers to the dominantly reversed polarity chron interval(s) separating the numbered anomaly (e.g. Anomaly 21) and the next older anomaly (e.g. Anomaly 22). The "Deep Sea Drilling Project" is abbreviated as DSDP.

PALEOMAGNETIC SAMPLING AND LABORATORY PROCEDURE

The sampling procedure used in this study is similar to standard paleomagnetic sampling techniques outlined by previous workers (e.g. McElhinny, 1973). At least three oriented samples were collected from every sampling site within the magnetostratigraphic sections. A minimum of three samples are required for analysis of the statistical significance of calculated mean paleomagnetic directions (Watson, 1956a, b; Irving, 1964; McElhinny, 1973). Lateral spacing between samples varied from 0.1 to 15 m. Oriented samples were collected from outcrop exposed beneath the superficial weathering layer, either as block hand samples or drilled cores. Cores were drilled with either of two portable, gasoline powered, water cooled drills, the 1-H.P. OR Engines "Drillgine" drill with Felker 1-in. internal diameter, 6-in. long diamond impregnated bits, or the 30.1 cc RocDrill with diamond-tipped bits. A custom-made core orientation sleeve was used to measure core azimuth and dip. Bedding strike and dip were also recorded to allow tilt correction of the magnetic vectors measured in the lab.

All sites were precisely located, both geographically and stratigraphically. The vertical stratigraphic separation of sites was determined by using some combination of hand level, tape measure, Brunton compass and clinometer, and Jacob's staff. I tried to maintain an average site separation of 3—7 m, although the nature of the exposed lithology, reconnaissance versus detailed sampling, location of reversal boundaries, etc. frequently resulted in greater or lesser distances between sites. For this entire study the average site spacing was approximately 1 site/9 m, with a total of 217 sampling sites over approximately 2000 m of stratigraphic section. The geographic location of all paleomagnetic stratigraphy sections and sampling sites were plotted on 7 1/2 minute, U.S.G.S. topographic quadrangle sheets. Precise geographic positions were determined by using distinctive topographic and geographic features, the intersection of multiple backsightings to features of known location, topographic benchmarks, altimeter readings, etc.

The finest-grained lithologic units were sampled whenever possible, with the majority of sites of claystone to fine grained sandstone lithology.

Paleomagnetic sections were preferentially sampled at published type stratigraphic sections to maximize stratigraphic utility. Several units were sampled 2—10 km from the type sections to allow better sampling or overlap of particularly important portions of the section. Physically continuous, distinct, traceable marker beds were used to tie together laterally separated segments of lengthy magnetostratigraphic sections. Factors emphasized in section selection included: stratigraphic continuity; overlap of physically traceable marker beds; areas of minimal faulting; regions of folding for applications of a fold test for stability; direct association with stratigraphic type sections, radioisotopically dated strata, faunal and/or floral assemblages; well exposed outcrop; availability of lithologically suitable strata (e.g. minimum grain size, relatively unweathered); sampling of a diversity of environments of deposition, paleoaltitudes, geographical areas, sediment source areas, tectonic settings, climatic zones etc.

In the laboratory, hand samples were preliminarily prepared and subdivided into approximately 3/4—1″ cubes using a stationary, circular disk sander, band saw with a metal cutting blade, or high speed, diamond-tipped circular saw. Cores were cut into approximately 3/4″ length by 1″ diameter cylinders using a double-bladed, parallel-mounted, diamond-tipped circular saw. Both cores and block specimens were hand sanded to remove surface impurities introduced during the sampling and coarse preparation procedures.

Laboratory analyses of the remanent magnetization of the samples were performed at the Paleomagnetics Laboratory, Lamont-Doherty Geological Observatory of Columbia University, Palisades, New York. All magnetization vector measurements were made on either a slow spin, Digico spinner magnetometer or a SC Technology superconducting cryogenic magnetometer. Some supplemental measurements and demagnetization studies have been performed at the Rutgers University Paleomagnetics Laboratory, using a Molspin Ltd. Minispin spinner magnetometer. Demagnetization of all samples was performed on a Schoenstedt GSD-1 single-axis AC demagnetizer, and a Schoenstedt TSD-1 thermal specimen demagnetizer. Susceptibility measurements were made on a Bison Instruments bulk susceptibility bridge. All specimens were transported from the demagnetizing equipment to the magnetometers in mu-metal shielding in order to prevent rapid acquisition of a viscous remanent magnetization in the laboratory. Helmholtz coils were used around both magnetometers at Lamont-Doherty to cancel the ambient field near the measurement areas.

Several terms that are used in the paleomagnetics discussions are defined here:

MDT — median demagnetizing treatment — demagnetization step or treatment at which 50% of the NRM intensity is removed.

x — magnetic susceptibility;

k — magnetic precision parameter;

G — Gauss;

oe — oersted;

NRM — natural remanent magnetization, measured prior to any demagnetization treatment;

J_{nrm} — magnetic *intencity* of the natural remanent magnetization;

primary magnetization — a stable, characteristic, magnetization believed to represent the magnetization acquired in the ambient geomagnetic field at the time (or very shortly after) a sample was deposited; a moderate to high coercivity and blocking temperature magnetization generally isolated during demagnetization to 200—600° or 625°C;

secondary magnetization — a magnetization believed to be acquired by a sample sometime *after* deposition, and therefore not representative of the geomagnetic field at the time of deposition of a sediment: generally either a very low or very high blocking temperature and coercivity component, that can be distinguished from the primary magnetization by use of vector demagnetization studies and magnetic stability tests.

GEOLOGIC SETTING

East Fork Basin area

The Eocene strata of the East Fork Basin area lie at the southern edge of the Absaroka Range in northwestern Wyoming. The range is formed by the generally horizontal andesitic sediments and igneous rocks of the Absaroka volcanic field.

Love et al. (1976) present radioisotopic data that indicate a shift in the timing of volcanism within the Absaroka—Gallatin volcanic province, beginning at about 54.9 Ma in the northwest and ending at about 38.8 Ma in the southeast. This same NW—SE transgression of volcanism was inferred by Smedes and Prostka (1972), based on Absaroka Volcanic Supergroup stratigraphy.

The reader is referred to Hague et al. (1899), Love (1939), Smedes and Prostka (1972), Bown (1982), Flynn (1983), and references cited in those papers for more information on the geology of the East Fork Basin and Absaroka Range areas.

In 1939, J.D. Love proposed and described the Eocene volcaniclastic strata of the Aycross, Tepee Trail and Wiggins Formations from the East Fork Basin area. The Aycross and Tepee Trail Formations are the principle units investigated in the present study.

The thickness of the Aycross Formation in this area ranges from 0 ft to a maximum of more than 1000 ft (300 m; at the type locality on North Mesa). In the East Fork Basin area Aycross Formation strata do not occur above a topographic elevation of around 8500 ft (2600 m). Love (1939) did not originally designate a type section because the physical appearance of the formation is extremely variable laterally. In the type area, basal Aycross Formation strata "consist of soft brilliantly variegated clays in which red, purple, and green dominate" (Love, 1939, p. 67), while the upper part of the

formation contains "variegated clays, shales, sandstones, conglomerates, and volcanic rocks" (Love, 1939, p. 67). This section contains several distinctive tuffs and conglomerates containing andesite roundstones in an andesitic tuff or microcrystalline matrix. Bentonites are variably present in the Aycross Formation in this area. Underlying rocks frequently control the lithology of the overlying sediments. Volcanic debris is a common constituent of the Aycross Formation.

The Tepee Trail Formation has the greatest areal distribution of any Cenozoic unit in the southeastern Absaroka Range, where it outcrops at elevariations between about 7700 ft and 10,500 ft (2347—3200 m; Love, 1939). The thickness of the Tepee Trail Formation ranges from a maximum of 1500—2000 ft (457—610 m; near the type section) to much lesser thicknesses in areas of great erosion or of deposition on a surface of high relief. At the type section the generally fluviatile and volcaniclastic Tepee Trail Formation is fossiliferous and is 1416 ft (431.5 m) thick, although its base is not exposed. The Tepee Creek paleomagnetic section was taken in excellent exposures just west of the type section and south of the Tepee Trail, from which the formation takes its name. Tepee Trail Formation lithologies at the type section and the Tepee Creek paleomagnetic section are volcanic tuff, conglomerate, siltstone, shale and clay with some minor breccia. Love (1939, p. 74 and 76) points out that most strata are well-bedded and are green or brown, lithologic components are almost exclusively basic andesitic material, intrusives are absent and flows are uncommon, the unit changes rapidly from fine-grained tuffs in the south and east to coarse breccias and conglomerates in the north and northwest, and individual beds are not very laterally continuous.

Love (1939) proposed the Wiggins Formation for 1000—3000 ft (305—914 m) of strata unconformably overlying the Tepee Trail Formation and older rocks in the East Fork Basin area. The unit rarely occurs below an elevation of 10,400 ft (3170 m), although just north of Crow Mountain it crops out as low as 9000 ft (2743 m).

A progressive increase in the abundance of volcanic detritus, and a tendency for the strata in this sequence progressively to coarsen-upwards is observed in the stratigraphic sequence from Aycross to Tepee Trail to Wiggins Formation. The Aycross Formation strata tend to be siltstones and mudstones with minor conglomeratic interbeds; the Tepee Trail Formation generally consists of tuffs, siltstones and sandstones with large numbers, and significant thicknesses, of conglomerates; while the Wiggins Formation consists of predominantly mudflow breccia, conglomerate and coarse-grained sandstone with minor amounts of finer sandstone, siltstone and lava flows.

Washakie Basin, Wyoming

The Eocene, fluviatile to lacustrine Washakie Formation is the youngest early Tertiary rock unit cropping out in the Washakie Basin of southwestern

Wyoming. The Washakie Basin is a synclinal, structural and topographic basin covering an area of approximately 2500 square miles (6500 square kilometers; Roehler, 1973). It is flanked by the Sierra Madre Range to the east, Wamsutter Arch to the north, Rock Springs Uplift to the west, and Cherokee Ridge to the south. Eocene strata in the basin are approximately 8000 ft (2438 m) thick. Surface sections of the Washakie Formation yield a maximum thickness of approximately 3000 ft (914 m), although Roehler (1969) indicated a possible subsurface thickness of 12,300 ft (3749 m) for the formation.

Roehler (1973) divided the Washakie Formation into a lower Kinney Rim Member and an upper Adobe Town Member. At its type section the Kinney Rim Member (Beds 515—568) is 893.9 ft thick (272.5 m; Roehler, 1973). A complete section for the Adobe Town Member (Beds 569—708) was compiled from the lower Adobe Town Member section (principal reference section; Beds 569—628, 883.8 ft (269.4 m) thick), the type section (middle Adobe Town Member, Beds 629—675, 1100.7 ft (335.5 m) thick), and the upper Adobe Town Member section (principal reference section; Beds 676—708, 341.0 ft (103.9 m) thick), yielding a total thickness for the member of 2325.5 ft (708.8 m; Roehler, 1973). The paleomagnetic sections in this study were all collected along these type and principal reference sections for the entire Washakie Formation.

Granger (1909) had earlier divided the "Washakie beds" or "Washakie Formation" of the Washakie Basin into the Lower Washakie (Horizon A — Beds 1—10) and Upper Washakie (Horizon B — Beds 11—22). Granger's Washakie A corresponds only to Roehler's (1973) lower Adobe Town Member, while Granger's Washakie B is equivalent only to Roehler's middle Adobe Town Member.

Mauger (1977) presented twenty-nine K—Ar dates (22 horizons) on Eocene strata from several western interior basins, including the Washakie Basin. Most of these results (20 of 29 dates) were considered by Mauger (1977) to be of low reliability for various reasons (argon loss, contamination by older detrital minerals, etc.). The wide variability in age determinations (24.0—153.2 Ma for Eocene strata), and the predominance of unreliable results (1 of 12 Washakie Basin results "reliable"), leads me presently to reject the one proposed reliable date from the Washakie Basin. For the same reasons, Mauger's (1977) dates have been excluded from Table VI.

Roehler (1973, p. 3—8) and Turnbull (1978, p. 573—585) provide a much more extensive review of the history of geologic and stratigraphic studies in the Washakie Basin. The reader should consult Roehler (1973, p. 23—39) for detailed lithologic descriptions and thicknesses of the individual beds (515—708) in his composite stratigraphic section of the Washakie Formation.

San Diego area, California

The Eocene strata of the San Diego area, Southern California provide an unusual geologic setting in which intertonguing, fossiliferous, Paleogene

marine and continental rocks are exposed in close geographic proximity. These Eocene strata unconformably overlie a basement complex of Jurassic to Cretaceous igneous and sedimentary rocks (see Kennedy, 1975). No Paleocene strata are known to crop out beneath Eocene strata in this area. Eocene rocks are the youngest Paleogene strata occurring in the San Diego area.

The Eocene strata in this area were deposited on a continuously subsiding narrow shelf, steep continental slope and continental margin basin (the San Diego embayment of Kennedy, 1975) that extended both south and north of what is today the San Diego metropolitan area. Strata analyzed in this study crop out in the Delmar, La Jolla, Poway and La Mesa 7 1/2 minute quadrangles. The Eocene strata around San Diego form an eastward thinning wedge of continental margin sediments (Lohmar and Warme, 1979), with a maximum thickness of approximately 700 m (Kennedy, 1975).

Early geologic work on the Cretaceous and Eocene strata in the San Diego area included studies by Ellis (1919), Hanna (1926), and Milow and Ennis (1961). Bukry and Kennedy (1969), Kennedy and Moore (1971), Peterson and Kennedy (1974), Kennedy (1975), and Kennedy and Peterson (1975) presented more recent studies.

Prior to 1970 the Eocene sediments were generally assigned to two stratigraphic units, the La Jolla Formation (of Clark, 1926; including Rose Canyon Shale Member of Hanna, 1926) and the Poway Conglomerate (of Ellis, 1919; redefined by Hanna, 1926). The La Jolla and Poway Groups (revisions of Kennedy and Moore, 1971; Kennedy and Peterson, 1975; see Fig.1.B for a diagrammatic stratigraphic profile) contain a sequence of nine intertonguing, partly laterally equivalent and contemporaneous formations.

The La Jolla Group generally lies stratigraphically below, and west of the Poway Group; it consists of deltaic conglomerate and sandstone, lagoonal sandstone and claystone, beach sandstone, marine shale, and non-marine strata (Kennedy, 1975). Included within the La Jolla Group are (from approximately bottom to top) the Mount Soledad Formation, Delmar Formation, Torrey Sandstone, Ardath Shale, Scripps Formation, and Friars Formation. The Poway Group strata include deltaic conglomerate and sandstone, lagoonal sandstone, littoral sandstone and siltstone, and non-marine strata (Kennedy, 1975). The Poway Group includes (from approximately bottom to top) the Stadium Conglomerate, Mission Valley Formation, and Pomerado Conglomerate.

Kennedy (1975, figs.5 and 6) and Lohmar and Warme (1979) portray the La Jolla Group as a transgressive/regressive cycle of three laterally accreting, laterally contemporaneous depositional environments. Sediments accumulated during this cycle as deep marine deposits (Ardath Shale), beach and nearshore marine shelf deposits (Torrey Sandstone and Scripps Formation), and lagoonal deposits (Delmar and Friars Formations). A second cycle of transgression/regression resulted in deposition of nearshore shelf deposits (Mission Valley Formation) and marine and continental deltaic deposits

(Stadium and Pomerado Conglomerates). Gastil and Higley (1977, fig.2) pointed out that parts of the mammal-bearing Friars and Mission Valley Formations are of non-marine origin.

In this study I follow the stratigraphic nomenclature of Kennedy and Moore (1971) and Peterson and Kennedy (1974). The work of Lohmar and Warme (1979) and Lohmar et al. (1979) provides an excellent treatment of the facies relationships and depositional environments of some of the San Diego Eocene strata.

BIOCHRONOLOGY

East Fork Basin area

Mammalian faunas from the type sections and areas for the Aycross and Tepee Trail Formations have been reported and discussed in several papers (Wood et al., 1936; Love, 1939; Rohrer and Obradovich, 1969; Lewis, 1973; Rose, 1978; Berggren et al., 1978; McKenna, 1980; MacFadden, 1980; Flynn and Galiano, 1982; Bown, 1982).

Two faunal horizons have been reported from the type area of the Aycross Formation on North Mesa (Wood et al., 1936). The lower locality (Horizon *A*; see Fig.9; including *Desmatotherium guyotii* = *Helaletes intermedius*, Radinsky, 1967; *Eotitanops borealis*; *Palaeosyops* cf. *major*; and *Patriofelis* sp.) lies more than 100 ft (30 m) below the upper locality (Horizon *B*, see Fig.9; including *Palaeosyops robustus*, *Patriofelis ferox*, *Hyrachyus eximius*, *H.* cf. *modestus* ("progressive variant"), *Uintatherium* cf. *mirabile*, *Telmatherium* cf. *cultridens*, *T.* cf. *validum*, and cf. *Tillotherium*). The fauna from the lower horizon is probably early Bridgerian in age, while the fauna from the upper Aycross horizon definitely is late Bridgerian in age (Wood et al., 1936; Bown, 1982). Wood et al. (1941) assigned a Bridgerian age to the Aycross Formation. Bown (1982) reported additional specimens from the Aycross Formation in its type area that do not further refine the temporal correlation of this unit.

Love (1939) reported a fragmentary composite fauna from three localities in the Tepee Trail Formation (including *Metarhinus* sp., similar to species from "Uinta B" age strata in southwestern Wyoming and Utah; and elements of unidentified species of titanothere, creodont, herbivore, rodent, crocodile, and one or more teleosts). Based upon this information Wood et al. (1941) assigned a Uintan age to the Tepee Trail Formation. Based on an incorrect taxonomic identification, Lewis (1973) suggested a possible Bridgerian age for this unit (see McKenna, 1980). Berggren et al. (1978), MacFadden (1980), and McKenna (1980) concluded that the extensive fauna now available from Unit 24 (Horizon D, see Fig.9) in the type section of the Tepee Trail Formation is early Uintan in age.

The fauna from Bone Bed A (Horizon *D*, see Fig.9) of the type section of the Tepee Trail Formation includes (in the definitions of West et al., in

press): (1) taxa with Uintan first appearances — *Oligoryctes*, *?Protadjidaumo typus* and another eomyid rodent, *Amynodon*, and selenodont artiodactyls; (2) Uintan index taxa — *Epihippus* (which is listed by MacFadden, 1980 as also occurring in the Duchesnean), *Amynodon advenus*, and *Achaenodon*; (3) characteristic Uintan taxa — *Epihippus* and *Amynodon*; (4) taxa with Uintan last occurrences — *Sciuravus* (and possibly *Ignacius*, which may have a Uintan last occurrence); (5) characteristic Bridgerian taxa — *Apatemys*, *Hyopsodus*, *Uintasorex*, *Herpetotherium* (=*Peratherium*), and *Sciuravus*; (6) taxa with Wasatchian first appearances — *Herpetotherium* (=*Peratherium*), *Nyctitherium*, and *Microparamys*; and (7) a taxon with a Wasatchian last occurrence — *Peradectes* (which Bown, 1982 records from the Bridgerian Aycross Formation from the eastern Absaroka Range). This fauna also includes cf. *Tetrapassalus* sp. (Rose, 1978); this rare genus is known only from the Bridgerian elsewhere.

On the species level, this Tepee Trail Formation type section fauna includes *Uintasorex parvulus* and *Trogolemur* cf. *T. myodes* (both known elsewhere only from the Bridgerian), *Hyopsodus* n. sp. (closely related to Bridgerian to early Uintan *H. paulus* and Uintan *H. uintensis*), *Epihippus uintensis* (Uintan to Duschesnean elsewhere), *Dilophodon minusculus* (Bridgerian elsewhere; although this could be the late Uintan species *D. leotanus*), and *Amynodon advenus* (Uintan elsewhere).

Washakie Basin

Paleontologists have collected actively in the Washakie Basin for more than 100 years, since the Hayden Surveys of the Western Territories began in 1867. The most important descriptions and summaries of the Washakie Formation are by Cope (1884), Granger (1909), Osborn (1929), Roehler (1973), and Turnbull (1972, 1978). The reader is referred to Roehler (1973) and Turnbull (1972, 1978) for references to the numerous papers that provided detailed descriptions and discussions of selective elements of the Washakie Formation faunas.

Granger (1909, fig.2 and p. 22) summarized the fauna of the Washakie Formation as then known. Of the 18 Washakie A genera listed by Granger, 3 are Bridgerian index taxa, 6 are characteristic Bridgerian taxa (four of these have Uintan last appearances), and one has a Bridgerian first appearance (ranges as in West et al., in press). Of Granger's (1909) 13 Washakie B genera, one is a Uintan index, 3 are characteristic Uintan taxa (one of these has a Uintan first appearance), and 6 have Uintan last appearances (two of these are early Uintan last appearances). Osborn (1929) added 4 new genera to the Washakie A fauna, none of which provided significant biostratigraphic information.

Roehler (1973) and Turnbull (1972) provided faunal lists from the Kinney Rim Member, lower Adobe Town Member, and middle Adobe Town Member of the Washakie Formation.

These faunas strongly support the Bridgerian age assignment for the lower Adobe Town Member (Horizon *A*, see Fig.9; Washakie A of Granger) and the Uintan age assignment for the middle Adobe Town Member (Horizons *C* and *D*, see Fig.9; Washakie B of Granger). In particular, the presence of Bridgerian index taxa (*Orohippus*, *Palaeosyops*, *Stylinodon*, and *Thinocyon*), a genus with a Bridgerian last occurrence (*Patriofelis*) and several characteristic Bridgerian taxa (*Hyopsodus*, *Miacis*, *Sciuravus*, *Paramys*, *Notharctus*, *Microsyops*, *Hyrachyus*, and *Orohippus*) indicate a Bridgerian age for the lower Adobe Town Member. West et al. (in press) also list *Notharctus robustior* and *Hemiacodon* (both considered "unique to late Bridgerian"), *Tetheopsis*, *Uintatherium* (Bridgerian to early Uintan range, "abundant" in late Bridgerian), *Mesatirhinus*, and *?Diplobunops* as evidence of a late Bridgerian age for the lower Adobe Town Member.

Similarly, the presence of Uintan index taxa (*Achaenodon*, *Amynodon*, and *Protoptychus*), several Uintan last occurrences (*Hyopsodus*, *Ischyrotomus*, and *Limnocyon*), characteristic Uintan taxa (*Amynodon*, and *Ischyrotomus*, and possibly *Protylopus* and *Eobasileus*) indicate a Uintan age for the middle Adobe Town Member. West et al. (in press) consider the additional presence of *Dolichorhinus*, *Eomoropus*, and *Triplopus* in the middle Adobe Town Member as evidence of its early Uintan age. An *early* Uintan age may further be suggested by the presence of *Protoptychus*, *Limnocyon*, *Metarhinus*, and possibly uintatheres. All of these taxa have early Uintan last occurrences (West et al., in press).

The fauna from the "rim below the Adobe Town Rim" (in the upper part of the lower Adobe Town Member; Horizon *B*, see Fig.9) is of equivocal age, containing *Notharctus robustior* (late Bridgerian index — but also found in San Diego), *Stylinodon* (late Bridgerian index taxon); *Hemiacodon* (characteristic Bridgerian genus, but also found in San Diego); *Paramys*, *Sciuravus nitidus*, and *Hyopsodus* (characteristic Bridgerian; Uintan last occurrence); *Miacis* (Bridgerian characteristic); *Uintatherium* (early Uintan last occurrence); and *Tetheopsis*, *Mesatirhinus*, (?) *Diplobunops*, and *Manteoceras*. Further microfaunal localities in the lower Adobe Town Member between the classic Bridgerian and Uintan faunas are currently under study by W. Turnbull; these may help determine the precise location of the Bridgerian/Uintan boundary and "earliest Uintan" time within this section.

San Diego area

Golz (1973), Golz and Lillegraven (1977), and Lillegraven (1979) provided summary information regarding the extensive mammalian fauna from the San Diego area. Other authors, including Stock (1937, 1939), Wilson (1940a—c), Hutchison (1971), Golz (1976), Novacek (1976), Schiebout (1977), and Lillegraven (1979, 1980) have provided more detailed descriptions of various elements of the San Diego area mammalian fauna.

The Ardath Shale and Scripps Formation produce a poorly preserved, tax-

onomically depauperate, and numerically limited mammalian assemblage (see Fig.9, Horizon *A*; Hutchison, 1971; Golz, 1976; Kennedy, 1975; Golz and Lillegraven, 1977). This assemblage tentatively has been assigned a Bridgerian or earliest Uintan (Uinta A) age (Kennedy, 1975; Golz, 1976), although such a poorly preserved assemblage can provide only a tenuous age assignment.

Most of the San Diego area mammals have been found in the Friars and Mission Valley formations (see Fig.9, Horizons *B* and *C*), as summarized in Lillegraven's (1979, tables 1 and 2) composite species list. This list is a composite of all the taxa reported from localities in strata of the La Jolla and Poway Groups. Lillegraven's (1979) list included species he considered (1) "endemic" to the West Coast, and "markedly distinct" from species of the Rocky Mountain area, or (2) "extremely similar to or conspecific with" species of the Rocky Mountain area. Based on these taxa Lillegraven (1979) considered the San Diego fauna as ?earliest Uintan in age. In Lillegraven's (1980) study of primates from the San Diego area, he considered the San Diego area fauna to be early Uintan in age.

The fauna from the Friars/Mission Valley Formations of San Diego (see Lillegraven, 1979) includes taxa (in the definitions of West et al., in press): (1) with a Uintan first appearance — *Amynodon*, *Simidectes*, *Protoreodon*, and *Leptoreodon*; (2) with a Uintan last occurrence — *Ischyrotomus*, *Metarhinus*, *Sciuravus*, and cf. *Uintatherium*; (3) that are characteristic of Uintan — *Protoreodon*, *Amynodon*, *Protylopus*, and *Ischyrotomus*; (4) that are characteristic of Bridgerian — *Microsyops*, *Notharctus*, *Omomys*, *Hemiacodon*, *Uintasorex*, *Sciuravus*, *Herpetotherium* (=*Peratherium*), and *Apatemys*; (5) that are Bridgerian index taxa — *Notharctus* sp. near *N. robustior* and *Microsyops* sp. cf. *M. annectens*; (6) with a Bridgerian first appearance — cf. *Harpagolestes*; (7) with a Wasatchian first appearance — *Leptotomus*, *Nyctitherium*, *Omomys*, and *Microparamys*; and (8) that is a Wasatchian index — *Pelycodus* sp. near *P. ralstoni*.

On the species level, this San Diego fauna includes *Amynodon reedi*, to which Wall (1982) has assigned several specimens from the Uinta B, Uinta Formation, Utah; *Protoreodon* sp. cf. *P. parvus*, which Golz (1976) considers closely related to *P. parvus* from the Uinta B—C, Uinta Formation, Utah; *Leptoreodon* sp. cf. *L. marshi*, which Golz (1976) considers conspecific with *L. marshi* from the Uinta B—C, Uinta Formation, Utah; and *Dilophodon leotanus*, which occurs in the late Uintan (Uinta C and Randlett, Utah) of the western interior (however, Schiebout, 1977, assigns this material to *Dilophodon* sp., which could represent the Bridgerian species *D. minisculus* — see Radinsky, 1963). The primates from this fauna include (Lillegraven, 1980) *Notharctus* sp. near *N. robustior*; *Pelycodus* sp., unnamed, near *P. ralstoni*; *Hemiacodon* sp. near *H. gracilis* (all three species are Bridgerian elsewhere); *Washakius woodringi* (close to Bridgerian *W. insignis*), *Microsyops* sp. cf. *M. annectens* (a Bridgerian species), and *?Macrotarsius* sp. near *M. jepseni* (species only known from the early Uintan of Utah).

A brief summary of the San Diego marine biostratigraphy is presented here.

Molluscan biostratigraphy indicates a correlation of the Delmar Formation (possibly), Ardath Shale, and lower and intermediate portions of the Scripps Formation with the "Domengine Stage". The upper part (possibly the upper half) of the Scripps Formation is correlative with the "Transition Stage". Kennedy (1975) considered a fauna from the lower Friars Formation (near the gradational boundary with the Scripps Formation) possibly equivalent to the "Transition Stage". Givens and Kennedy (1979) reported a small assemblage from the upper Friars Formation that is characteristic of the "Transition Stage", but they also stated that stratigraphic relations indicate that the lower Friars Formation may be equivalent to the "Transition Stage" and also, in part, to the "Domengine Stage", while the upper Friars Formation may be equivalent to the "Transition Stage" or "Tejon Stage". The Stadium Conglomerate may be correlative with the "Tejon Stage", while the Mission Valley Formation is equivalent to the "Tejon Stage" (or possibly only the *upper* part of the "Tejon Stage").

Clark and Vokes (1936) considered the "Domengine Stage" to be equivalent to the Lutetian Stage from the Paris Basin, based on 12 closely related species pairs shared between the two regions. Clark and Vokes (1936) *only* used stratigraphic position to infer that the "Transition Stage" is correlated with the uppermost Lutetian or the Auversian Stage of the Paris Basin. Although only a few taxa are shared between California and Europe, Clark and Vokes (1936) inferred a correlation of the "Tejon Stage" with the Auversian or Bartonian Stage. Givens (1974) considered the "Tejon Stage" correlative with the upper Eocene of Europe, whereas Givens and Kennedy (1979) inferred a late middle Eocene (late Lutetian) age for the "Tejon Stage".

Detailed analysis of the benthonic foraminifera from the San Diego area indicates that the strata sampled in this study are assignable to the Ulatisian and Narizian. The foraminifers from the Ardath Shale appear to belong entirely within the Ulatisian Stage (Gibson and Steineck, 1972), although the Ulatisian/Narizian boundary *may* lie within, or slightly above, the uppermost Ardath Shale (Mallory, 1959; Phillips, 1972). Narizian benthonic foraminifers occur within the Stadium Conglomerate, and possibly within the lower Mission Valley Formation (associated with a "Tejon Stage" molluscan assemblage).

Gibson (1976) considered the benthonic foraminiferal zones to be time-transgressive, facies and depth controlled associations, based on comparisons with planktonic foram and calcareous nannoplankton temporal zonations. He stated that Californian Ulatisian faunas range in age from early Eocene (Ypresian) to early middle Eocene (Lutetian), while Narizian assemblages are entirely middle Eocene (Lutetian) in age. Gibson's (1976, p. 103) biostratigraphic zonation chart showed Ulatisian ranging from the base of Zone P6 to the base of Zone P10, and from the base of the *Discoaster diastypus* Zone to the top of the *D. sublodoensis* Zone. The Ulatisian/Narizian boundary lies

somewhere within Zone P9 and the *Rhabdosphaera inflata* Subzone of the *D. sublodoensis* calcareous nannoplankton zone. Poore (1980, fig.2) showed the Ulatisian/Narizian boundary lying somewhere between the middle of the *D. sublodoensis* Zone (NP14) to the lower half of the *Nannotetrina quadrata* (approximately NP15) Zone, and within Zone P10. Givens and Kennedy (1979) stated that the middle/late Eocene boundary on the Pacific Coast of North America lies near the Narizian/Refugian Stage boundary. Neither this boundary, nor the Refugian Stage, have been recognized within the San Diego Eocene section.

Extensive planktonic foraminifer faunas occur within the Ardath Shale. These faunas are clearly correlative with, but probably not assignable to, the *Hantkenina aragonensis* (P10) and/or *Globigerapsis kugleri* (P11) Zone of the standard planktonic foraminiferal zonation. Most marine biostratigraphers draw the early/medial Eocene boundary at the base of Zone P10 (see Berggren et al., 1978; Poore, 1980; Berggren et al., 1985), so the Ardath Shale is certainly early middle Eocene in age.

The Stadium Conglomerate contains a sparse planktonic foraminiferal fauna. Gibson (1971), Steineck and Gibson (1971), Gibson and Steineck (1972) and Steineck et al. (1972) considered this assemblage to be late middle Eocene in age, based almost exclusively on the joint occurrence of *Globorotaloides suteri* and *Truncorotaloides collacteus*. Gibson (1971) considered this poor assemblage to represent a *Globorotaloides suteri* fauna, possibly correlative to the *Orbulinoides beckmanni* Zone (Zone P13). Givens and Kennedy (1979), however, believed that the joint occurrence of these two species was indicative only of a broad late middle Eocene (later Lutetian) to late Eocene age (Bartonian—Priabonian age; although many workers place Bartonian within the middle Eocene).

Calcareous nannofossil floras from the Ardath Shale are definitely assignable to the *Rhabdosphaera inflata* Subzone of the *Discoaster sublodoensis* (NP14) Zone. In the type section of the Ardath Shale, localities within the *D. sublodoensis* Zone lie at the same horizon as "Domengine" mollusks (Kennedy, 1975). Gibson (1976, p. 103) considered Zone NP14 to be equivalent to Zone P9, whereas Poore (1980) indicated that the base of middle Eocene (base Zone P10) coincides with the base of Zone NP14. Berggren et al. (1985), however, consider the base of the middle Eocene (base Zone P10) to lie within Zone NP14.

Only sparse nannofloras of equivocal age have been reported from the Poway Group strata. Depauperate nannofloras of uncertain age have been reported by Kennedy and Moore (1971) and Kennedy (1975) from the Stadium Conglomerate. Kennedy (1975) and Givens and Kennedy (1979) considered a sparse flora from the Mission Valley Formation to be indicative of a late middle and/or early late Eocene age. Bukry (1980) assigned an "upper upper Eocene" "age" to the calcareous nannofossil flora from the Mission Valley Formation; no justification was given for this unusual assignment. The sparse assemblages of planktonic forams and calcareous nanno-

fossils from the Stadium Conglomerate and Mission Valley Formation provide only a most tenuous age correlation (see Fig.9) for the upper part of the San Diego area sequence.

The most definite biostratigraphic age assignments in the San Diego area Eocene sections have been for the faunas and floras of the Ardath Shale. The Ardath Shale contains floras assignable to the *Rhabdosphaera inflata* Subzone of the *Discoaster sublodoensis* Zone (upper part of Zone NP14), planktonic foram faunas correlative with the *Hantkenina aragonensis* (Zone P10) and/or *Globigerapsis kugleri* Zone (Zone P11; equal to the *Globigerinatheka subconglobata* Zone), and benthonic foraminifera assignable to the *Amphimorphina californica* Zone of the Ulatisian Stage. The Ulatisian/Narizian boundary coincides approximately with the Zone NP14/15 boundary and it lies within the top of, or slightly above, the Ardath Shale. This supports the upper Zone NP14 age assignment for the Ardath Shale. As the early/middle Eocene boundary (Ypresian/Lutetian Age boundary) is defined as the base of the *H. aragonensis* (Zone P10) planktonic foraminifer zone (and is variably correlated with the base or the middle of the *D. sublodoensis*, Zone NP14 calcareous nannofossil zone), the early/middle Eocene boundary *must* lie within the lowest part of, or more likely below, the Ardath Shale.

PALEOMAGNETIC SECTIONS — LOCATIONS AND DESCRIPTIONS

East Fork Basin area, Wyoming

Tepee Creek section

Paleomagnetic sampling information for the Tepee Creek (Tepee Trail Formation) paleomagnetic section is given in Table I. The type section of the Tepee Trail Formation lies 1 km (1/2 mile) due east of the base of the Tepee Creek section, and easily can be correlated stratigraphically to the Tepee Creek section.

The general lithology of the Tepee Trail Formation sediments at this section includes volcaniclastic, tuffaceous siltstones, sandstones and conglomerates. Sampling generally was limited to the siltstone to fine-grained sandstone lithologies. Most of the strata in this sequence weather to shades of blue, green, and brown, and are shades of gray, blue, or green on unweathered surfaces. There are no pink, red, orange or similarly colored units in this sequence that might suggest the obvious presence of hematitic staining and coloration. One fossiliferous locality, Tepee Creek No. 2, occurs directly within this paleomagnetic section; Bone Bed A in the type section can be correlated easily into this section.

Aycross Formation section

Sampling information for the Aycross Formation paleomagnetic section is given in Table I.

The Aycross Formation sediments are generally fine-grained, and the lith-

TABLE I

Geographic, stratigraphic and sampling information for the paleomagnetic sections of this study

Section	7 1/2 minute Quadrangle Map	Geographic location of section	No. of sites	No. of samples	Stratigraphic thickness	Bedding strike	Bedding dip	Stratigraphic units sampled
Tepee Creek	East Fork Basin, Wyoming	NW 1/4 Sec.4, NE 1/4 Sec.5 (T.43 N., R.104 W.) SW 1/4 Sec.33, SE 1/4 Sec.32 (T.44 N., R.104 W.)	31	101	939 ft (286 m)	315°	2°NE	Tepee Trail Fm.; along banks of Tepee Creek, East Fork Basin, base and top of fm. not sampled (poor exposure)
Aycross Formation	Bain Draw and Crow Mountain, Wyoming	NE 1/4 Sec.7, S 1/2 Sec.17, (T.7 N., R.5 W.)	39	121	1500 ft (457 m)	65° to 145°	6°SW to 11.5°SW to 6.5°SE	Aycross Fm.; potential type section; none yet designated; Badlands, between East Fork Wind River and N-end North Mesa
Kinney Rim Member	Chicken Creek East, Wyoming	S 1/2 Sec.16, SW 1/4 Sec.15, (T.14 N., R.99 W.)	27	77 (+28 sub-samples)	1395 ft (425 m)	345°	20°NE	Kinney Rim Mbr. (type section; Beds 515—568), Washakie Fm.; E side Kinney Rim, on S slopes of E—W drainage
Adobe Town Member	Cow Creek Reservoir SW, Wyoming—Colorado	NW 1/4 Sec.29, SW 1/4 Sec.30, (T.13N., R.98 W.)	34	106 (+11)	862 ft (263 m)	315°	13°NE	Adobe Town Mbr. (lower; Beds 569—628), Washakie Fm.
Skull Creek	Upper Powder Spring, Wyoming—Colorado	N 1/2 Sec. 29, W 1/2 Sec.20, W 1/2 Sec.17, NE 1/4 Sec.18, (T.13N., R.97 W.)	24	80 (+11)	1202.5 ft (366.5 m)	280°	7°NE	Adobe Town Mbr. (middle; Beds 569—628), Washakie Fm.; Slopes Skull Crk, onto Skull Crk Rim

Delmar Formation	Delmar, California	S 1/4 Sec.1, W 1/2 Sec.11, S 1/2 Sec.11, E 1/2 Sec.23, (T.14S., R.4 W.)	11	35	approximately 150 ft (46 m)	Delmar Fm. (type section), La Jolla Grp.; Along sea cliffs 2 km S of Delmar RR Station
Ardath Shale Type	La Jolla, California	unsurveyed — SW 1/4 (T.15S., R.3 W.)	7	22	approximately 150 ft (46 m)	Ardath Shale (type section), La Jolla Grp.; E side Rose Canyon; 800 m S of Ardath Rd./I-5
Scripps Formation Type	Delmar, California	unsurveyed — NE 1/4 (T.15S., R.4 W.)	16	47	400 ft (122 m)	Ardath Shale and Scripps Fm. (type section), La Jolla Grp.; sea cliffs, N side Black's Canyon
Type Friars Formation	La Jolla, California	unsurveyed — SE 1/4 (T.16S., R.3 W.)	6	18	125 ft (38 m)	Scripps and Friars (type section) Fms.; La Jolla Grp.; N wall Mission Valley, below San Diego U. High Sch.
Genessee Avenue	La Jolla, California	unsurveyed — NE 1/4 (T.16S., R.3 W.)	5	20	66 ft (20 m)	Scripps and Friars Fms., La Jolla Grp.; SSE facing hill, along Genessee Ave., SW of S.D. Mesa Coll.
Murphy Canyon	LaMesa, California	unsurveyed — NW 1/4 (T.16S., R.2 W.)	11	39	approximately 100 ft (30.5 m)	Friars Fm., La Jolla Grp. and ?Stadium Cgl., Poway Grp.; Murphy Canyon, along I-15, N of Stadium
Mission Valley Formation (3 locations)	Poway, California	(a) S 1/2 Sec.33, NE 1/4 Sec.33, (T.14S., R.2 W.) (b) N 1/2 Sec.34, NE 1/4 Sec.34, (T.14S., R.2 W.)	8	31	approximately 100 ft (30.5 m)	Mission Valley Formation, Poway Grp. (a) S of Miramar Reservoir (b) S of Miramar Reservoir
	LaMesa, California	(c) unsurveyed-SW 1/4, (T.16S., R.2 W.)				(c) W side of canyon, near intersection Fairmont Av/ Montezuma Rd

ologies sampled in the section range from claystone to fine-grained sandstone. Siltstone is the dominant lithology sampled in this section, followed by sandy siltstones. These strata are largely fluviatile in origin. Most of the strata sampled in this section weather to shades of red, purple, gray, and green, with minor yellow and brown horizons. On fresh surfaces these sampled units are predominantly reddish or green, with lesser numbers of gray, brown, purple or yellow samples. The red and purple coloration of some of these strata suggest that a hematite pigmentation may be present. The two fossiliferous Aycross Formation horizons of Wood et al. (1936, see above) probably occur directly within this paleomagnetic section.

Washakie Basin, Wyoming

Kinney Rim Member section

Paleomagnetic sampling information for the Kinney Rim Member paleomagnetic section is given in Table I. The base of the section is approximately 100 ft (30 m) below the "white ridge marker bed" of Roehler (1973, Bed 515), which marks the conformable, arbitrary contact between the Kinney Rim Member (Washakie Formation) and the Laney Shale Member of the Green River Formation. The top of this section extends above the "lower brown sandstones" (Bed 569 of Roehler, 1973; base of this unit marks the Kinney Rim Member/Adobe Town Member contact). These "lower brown sandstones" overlie a basin-wide unconformity, that is only recognizable on small-scale aerial photographs by the thinning and wedging-out (by truncation) of underlying beds against the "lower brown sandstones" (Roehler, 1973). Both the "lower brown sandstones" and overlying units and marker beds at the base of the Adobe Town Member can be physically traced laterally between the Kinney Rim Member and the Adobe Town Member paleomagnetic sections. Faunal locality FM 4-73 (Turnbull, 1978) lies directly within this section at approximately bed 517-533 (W. Turnbull, personal communication, 1980) and locality FM 11-70 (Turnbull, 1978; approximately Bed 526-540, W. Turnbull, personal communication, 1980; including locality No. 1 of Roehler, 1973 — Bed 527) is found nearby. Numerous other faunal localities (listed in Roehler, 1973; Turnbull, 1978) from around the basin can be correlated easily into all the Washakie Formation sections.

The Kinney Rim Member strata are predominantly fluviatile in origin, although there are several units within this section that may be of lacustrine origin. Roehler (1973, p. 13) states that the Kinney Rim Member is "composed mainly of gray, green, and some red mudstone and interbedded gray and gray-green, very fine to fine grained sandstone; and occasional thin beds of gray limestone, algal limestone, gray limy siltstone, and light-gray to white tuff." The strata sampled in this section generally weather to gray, brown, or green, although a few units weather to tan or yellow. On fresh surfaces these strata are predominantly gray, brown or green with minor numbers of tan, buff or yellow samples. Lithologies sampled for paleomagnetic analyses in-

clude siltstones to fine-grained sandstones, limestones, and occasional supplementary medium-grained sandstone samples.

Adobe Town Member section

Paleomagnetic sampling information for the Adobe Town Member paleomagnetic section is given in Table I. The base of the section is in the "lower brown sandstone" (Bed 569 of Roehler, 1973). The top of the section lies at approximately Beds 626—627 of Roehler (1973).

Sediments of the Adobe Town Member are predominantly fluviatile in origin, although there are some lacustrine interbeds. The general lithology of the Adobe Town Member is "...green, gray and red tuffaceous mudstone alternating with gray fine- to coarse-grained tuffaceous and arkosic sandstone and minor thin beds of green shale, light-gray and green tuff, gray siltstone, and conglomerate" (Roehler, 1973, p. 15). The sediments sampled for paleomagnetic analyses include limestones, claystones, siltstones, and fine-grained sandstones, although medium-grained sandstones were sometimes sampled for potential supplemental information from otherwise barren intervals in coarse-grained lithologies. Most of these sediments weather to gray, green, yellow or brown colors, although a few horizons weather to shades of white, tan, pink or red. Fresh rock surfaces are generally gray, green, or brown, with minor numbers of buff, yellow or red samples.

Skull Creek section

Paleomagnetic sampling information for the Skull Creek paleomagnetic section is given in Table I. Faunal locality FM 6-69 of Turnbull (1978) lies within this paleomagnetic section, at approximately Roehler's (1973) bed 635-644 (W. Turnbull, personal communication, 1980).

There are conformable contacts between the middle Adobe Town Member (type section) and both the upper and lower parts of the Adobe Town Member. Three marker units (Beds 616—620, 626—628, and 630) were physically traced laterally from the Adobe Town Member section approximately 10 km east to the Skull Creek section. The top of the Adobe Town Member paleomagnetic section (Beds 626—628) can be traced into the Skull Creek section approximately 45 m (140 ft) above its base. The top of the Skull Creek paleomagnetic section coincides with the top of Roehler's (1973) Adobe Town Member stratigraphic type section, which lies at Bed 675.

The Skull Creek section deposits are predominantly, or entirely, fluviatile in origin, as described in the lower Adobe Town Member section. Strata sampled in this section predominantly weather to gray, brown, or green with minor beds that weather to buff or yellow. Unweathered surfaces on these strata are generally gray or green, although some are white, yellow, brown, or red. The majority of the units sampled for paleomagnetic analyses were siltstones. Minor numbers of very fine- to fine-grained sandstone, some limestone, and a few supplementary medium-grained sandstone units were also sampled.

San Diego area, California

Seven paleomagnetic sections were sampled in the San Diego area. Most of these sections coincide with type sections (see Kennedy and Moore, 1971) of formations within the La Jolla and Poway Groups. A total of 203 samples from 62 sites were collected and analyzed from these sections. The total stratigraphic thickness covered by these sampling sites is greater than 335 m (1100 ft). Paleomagnetic sampling and geographic information for these sections is given in Table I.

The lithologies of the San Diego area sections vary widely depending on their environments of deposition. Lithologies range from mudstones to fine-grained sandstones, with some limestones, calcareous clastics, and some medium-grained sandstones. Volcanic input is negligible. Inferred environments of deposition range from bathyal to shallow water shelf, nearshore, subaqueous delta or subaerial alluvial fan, to fluviatile. Sampling was limited to the mudstone to fine-grained sandstone lithologies in most cases. In general, the strata in these sequences weather to gray, green, brown, or yellow, with some white and reddish strata. Unweathered surfaces are generally green, gray, brown, blue, or occasionally reddish. The samples from the Delmar Formation are almost entirely gray-green to dark green (green to green-blue on fresh surfaces) siltstones and sandy siltstones, with the samples from one site consisting of interbedded claystone and fine-grained sandstone. Gray and brown sandy siltstones dominate the Ardath Shale Type Section samples, although fine-grained sandstones also are common. The Scripps Formation Type Section samples are predominantly siltstones to fine-grained sandstones that are gray, brown, and yellow. The Type Section of the Friars Formation consists of gray to brown siltstones at the base (Scripps Formation) to fine to medium-grained sandstones (Friars Formation). The Genessee Avenue (Scripps and Friars Formations) lithologies range from claystones to fine-grained sandstones, and bedding colorations include green, white, and brown. Gray, green, and brown siltstones dominate the Murphy Canyon section (Friars Formation to ?Stadium Conglomerate), and claystones and fine-grained sandstones are present in lesser amounts. The sampled Mission Valley Formation lithologies range from claystones to fine-grained sandstones; gray-green siltstones and sandstones predominate. Two Mission Valley Formation horizons are reddish to reddish-brown in color.

Numerous fossiliferous horizons occur within the sampled paleomagnetic sections. These include horizons containing mollusks, benthonic and planktonic foraminifera, calcareous nannoplankton, and mammals.

PALEOMAGNETIC RESULTS

NRM intensities of magnetization

NRM intensities of magnetization for 220 samples from the East Fork Basin area, for 252 samples from the Washakie Basin, and for 203 samples

TABLE II

Paleomagnetic data summaries for the paleomagnetic sections of this study. Intensity units in section A are in Gauss

	Tepee Trail	Aycross	E.F. Combined	Kinney Rim	Adobe Town	Skull Creek	Washakie Comb.	San Diego Comb.
NRM intensity								
(No. of samples)								
(1) range	6.47×10^{-6} to 4.54×10^{-3} (94)	1.60×10^{-7} to 1.94×10^{-4} (121)	1.60×10^{-7} to 4.54×10^{-3} (200)	1.30×10^{-7} to 2.76×10^{-3} (79)	6.9×10^{-8} to 4.84×10^{-4} (111)	4.30×10^{-7} to 1.69×10^{-5} (62)	6.9×10^{-8} to 2.76×10^{-3} (252)	8.00×10^{-8} to 1.38×10^{-4} (203)
(2) mean	1.56×10^{-4}	3.07×10^{-5}	8.69×10^{-5}	1.16×10^{-4}	9.25×10^{-5}	7.5×10^{-6}	8.42×10^{-5}	8.77×10^{-6}
(3) median	7.5×10^{-5}	1.35×10^{-5}	7.5×10^{-5}	1.35×10^{-5}	1.35×10^{-5}		1.35×10^{-5}	2.38×10^{-6}
Magnetic susceptibility	1.82×10^{-5} to 7.15×10^{-4} (16)	1.73×10^{-6} to 1.64×10^{-4} (51)	1.73×10^{-6} to 7.15×10^{-4} (67)	— —	4.63×10^{-6} to 5.03×10^{-4} (20)	— —	4.63×10^{-6} to 5.03×10^{-6} (20)	1.34×10^{-6} to 1.55×10^{-4} (65)
J_{nrm}/χ	0.12—1.55 (16)	0.05 — 10.42 (51)	0.05 — 10.42 (67)		0.02 — 2.26 (20)		0.02—2.26 (20)	0.02 — 3.96 (65)
Intensity behavior								
(1) increase	58/83 (70%)	68/121 (56%)	126/204 (62%)	59/77 (77%)	61/84 (73%)	33/46 (72%)	153/207 (74%)	88/155 (57%)
(2) % of (1) with R primary mag.	44/58 (78%)	51/68 (75%)	95/126 (75%)	47/59 (80%)	39/61 (64%)	19/33 (58%)	106/153 (69%)	45/88 (51%)
(3) major direction change with increase?								
N samples	few	approx. 1/2	approx. 1/4	few	none	1 only	few	approx. 1/4
R samples	most	approx. 60%	most	many	many	many	many	approx. 1/2
(4) % R sites in section	20/25 (80%)	18/36 (50%)	38/61 (62%)	17/22 (77%)	15/24 (63%)	4/13 (31%)	36/59 (61%)	18/49 (37%)
MDT								
Pilot samples	100—300°C 12/16 560°C 1/16 multiple 1/16				200°C 27/32 400—660°C 5/32 200 oe 17/23			450°C 5/10 600°C 5/10
Other samples	(83 total) 200 oe or 200°C 63 300—650°C or 300—500 oe 19 multiple 1	(116 total) 100 oe 16 300 oe 33 500 oe 10 200—530°C 21 560°C 21		(77 total) 100 oe 45 100—300 oe 11 600°C 1 multiple 7	(89 total) 200 oe or 200°C 80 400—660°C or 300—600 oe 9	(53 total) 100 oe 14 300 oe or 200°C 25 300—500°C 5 560°C 2 multiple 3		(155 total) 100 oe 8 400 oe 11 1000 oe 3 200°C 8 560°C 63 600°C 34 multiple 25

from the seven San Diego area sections are given in Table II (also see figures and discussion in Flynn, 1983). The significantly greater NRM intensities of the East Fork Basin area strata probably reflect the volcanogenic origin, and greater concentrations of ferromagnetic minerals, of these sediments. The lower intensities in the Washakie Basin probably are due to the fluvial-lacustrine origin of the sediments, and greater dilution of ferromagnetic components because of greater distance from volcanic sources. The NRM intensities in the San Diego area are significantly lower than intensities from both other areas, as would be expected in nearshore and deeper water marine sediments, and in continental deposits far removed from a significant source (e.g. volcanic) of detrital ferromagnetic minerals.

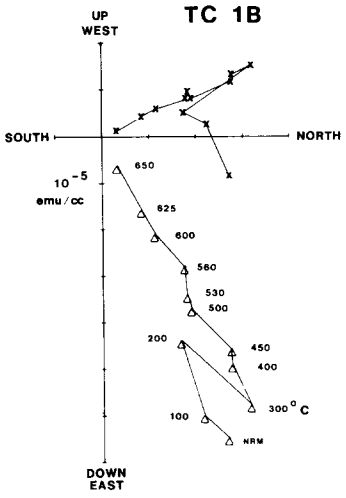
Demagnetization studies

East Fork Basin area

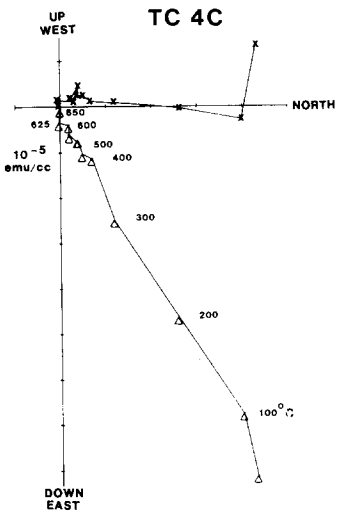
Detailed stepwise thermal demagnetization studies were performed on sixteen pilot samples from sixteen sites in the Tepee Creek section (greater than 50% of the total sites). Several of these studies revealed complex, multi-component magnetizations. Representative plots of vector behavior during progressive thermal demagnetization studies are shown in Fig.2. The remaining samples from this section therefore were treated with a combination of alternating field and thermal demagnetization. Every sample was demagnetized in at least four steps in an attempt to isolate various magnetic components with differing coercivity and/or blocking temperature spectra.

Samples 1B and 4C exhibit behavior characteristic of normally magnetized rocks in this section; a moderate to large secondary magnetization is removed by 300°C, and straight line decay of the vector, to the origin, occurs between 300 and 650°C. The primary vector in both cases is directed approximately to the north and down, as expected for a normal polarity sample from this latitude. Sample 1B exhibits a slightly more complex magnetization than 4C (especially in the interval from 300 to 560°C), but the secondary component is much weaker in sample 1B. The vector demagnetization plots for samples 10A, 11C and 31C illustrate the complex magnetizations characteristic of reversely magnetized rocks in this section. Strong secondary magnetizations are not completely removed until 300—450°C in these samples. All three show a large intensity drop between NRM and 100—300°C, followed by an increase in intensity to 450—530°C, and intensity decay above these temperatures to less than 10% of NRM at 600—650°C. It is important to note

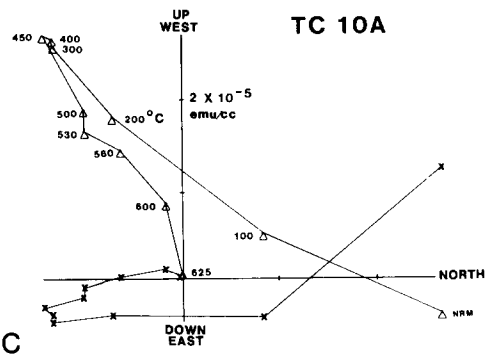
Fig.2. Vector demagnetization diagrams of samples from the Tepee Creek Section, East Fork Basin, Wyoming. Plots (after Zijderveld, 1967) of magnetization vector behavior during progressive thermal demagnetization from NRM to 650°C. Normal polarity samples TC 1B (A) and TC 4C (B); reverse polarity samples TC 10A (C), TC 11C (D), and TC 31C (E). X = horizontal component of vector, Δ = vertical component of vector, north is to right of page along horizontal axis, east and down is to bottom of page along vertical axis. Scale divisions are in 10^{-5} G (emu/cc).


TC 1B
UP
WEST
SOUTH
NORTH
10^{-5}
emu/cc
650
625
600
560
530
500
200
450
400
300°C
100
NRM
DOWN
EAST

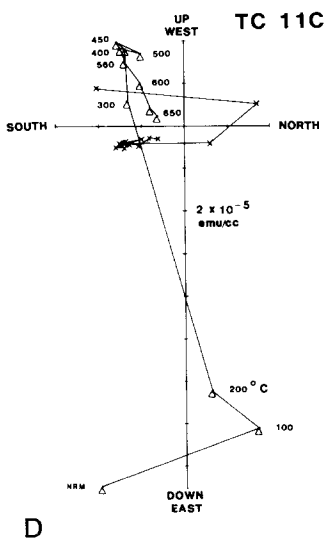
A


TC 4C
UP
WEST
NORTH
650
625
600
500
400
10^{-5}
emu/cc
300
200
100°C
DOWN
EAST

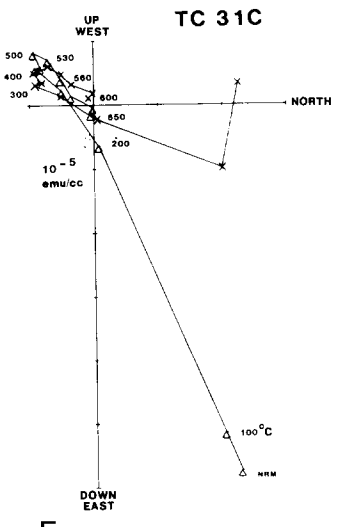
B


TC 10A
UP
WEST
450
400
300
500
200°C
530
560
600
100
625
NORTH
NRM
2×10^{-5}
emu/cc
DOWN
EAST

C


TC 11C
UP
WEST
450
400
560
500
600
300
650
SOUTH
NORTH
2×10^{-5}
emu/cc
200°C
100
NRM
DOWN
EAST

D


TC 31C
UP
WEST
500
530
400
560
300
600
NORTH
650
200
10^{-5}
emu/cc
100°C
NRM
DOWN
EAST

E

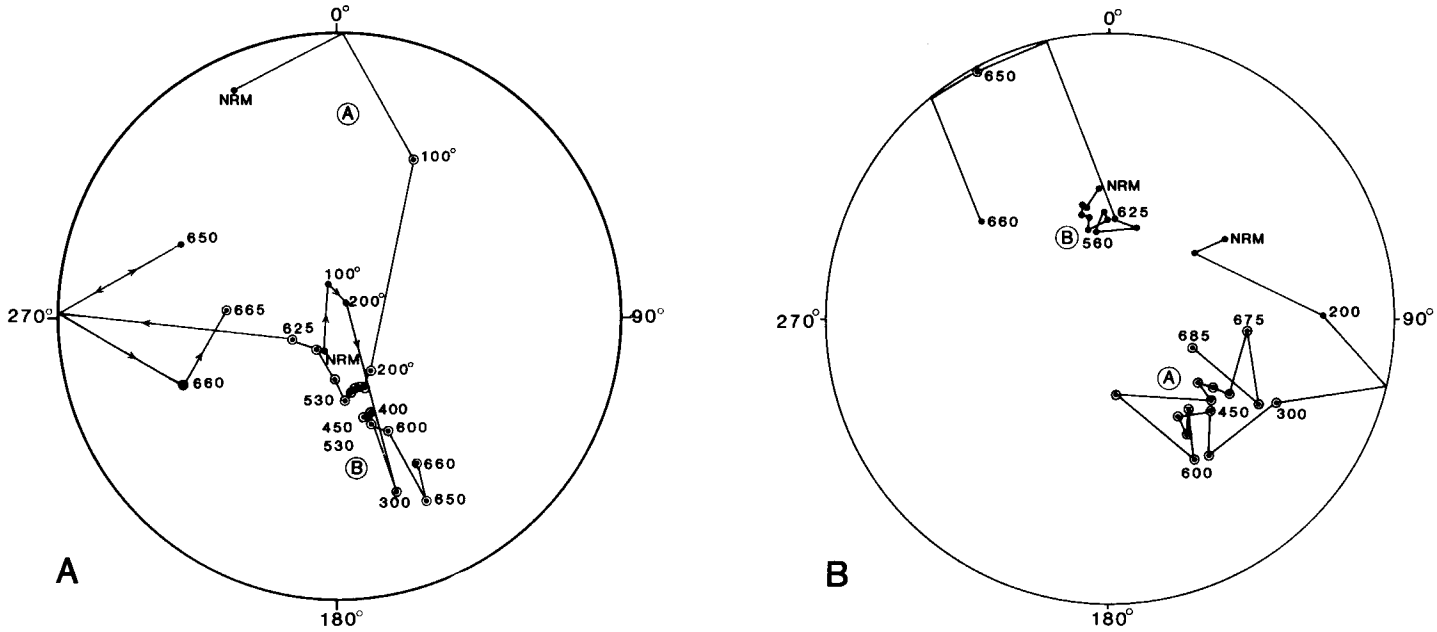


Fig.3. Stereographic projection. Magnetization vector directions during progressive thermal demagnetization from NRM to 665°C. A. reverse polarity samples A— TC 10A and B— TC 11C from the Tepee Creek section, East Fork Basin area. B. reverse polarity sample A— AM $18C_2$ and normal polarity sample B— AM 24B from the Adobe Town Member section, Washakie Basin. Open circles are negative inclinations (directed upwards, plot on upper hemisphere), closed circles are positive inclinations (directed downwards, plot on lower hemisphere).

that the characteristic, or primary, magnetization in sample 31C is not isolated until at least 400°C at an intensity less than 20% of the NRM intensity. Many samples in this section possess stable primary magnetizations at intensity levels of 10—25% of NRM intensities. In most samples, magnetizations with blocking temperatures above 650°C are random or unstable, although some preserve a stable hematitic component even above 650°C. Stereonet plots of some of these samples are shown in Fig.3. In particular, the stereonet plot for sample 10A shows a small, but consistent, directional change emphasized between 300 and 600°C; the vector diagram incorporates additional intensity information that indicates this change is only of minor importance in defining the stable, primary magnetization.

Major shifts in the direction of the magnetic vectors, to a stable orientation directed up and generally to the south, coincide with intensity decrease and subsequent increase (see Intensity Behavior, Table II). Such behavior would be expected in reversely magnetized rocks with a strong secondary magnetization aligned antiparallel to the reversed vector. Secondary magnetizations in the same direction as the present day field would be in the same direction as, and would reinforce, the primary magnetization of a normal polarity sample acquired in the closely similar Eocene paleomagnetic field. The more complex magnetizations present in the reversed samples may be caused by these antiparallel secondary magnetizations. Plots of remanent magnetization/NRM magnetization (J_R/J_0; see Flynn, 1983, figs.5.6 and 8.6, and discussions) during progressive thermal demagnetization of samples from both the East Fork Basin and Washakie Basin support these observations. Normal polarity samples generally exhibit steady intensity decay, while reverse polarity samples tend to yield intensity decreases followed by intensity increases during removal of an antiparallel secondary magnetization.

All of the samples in the Aycross Formation paleomagnetic section were demagnetized in a sequential series of three alternating field and six thermal demagnetization steps because of the complex magnetizations present in the samples from the Tepee Creek section. In general, the magnetizations of the samples from the Aycross Formation were less complex than those in the Tepee Creek section samples. Consistent remanent directions, and the characteristic (primary) magnetizations, were relatively easy to isolate in the Aycross Formation samples. Magnetization vectors, and vector decay during sequential demagnetization, were very stable in these samples.

Washakie Formation

Detailed progressive thermal demagnetization studies were performed on twenty-eight samples from twenty-eight sites (equal to about 33% of the sites in this formation). Stepwise alternating field demagnetization curves (sometimes followed by additional thermal demagnetization steps) were prepared for twenty samples from twenty sites. Most of these samples come from the same sites as samples used for thermal demagnetization pilot studies. Complex, multicomponent magnetizations with variable blocking temperature

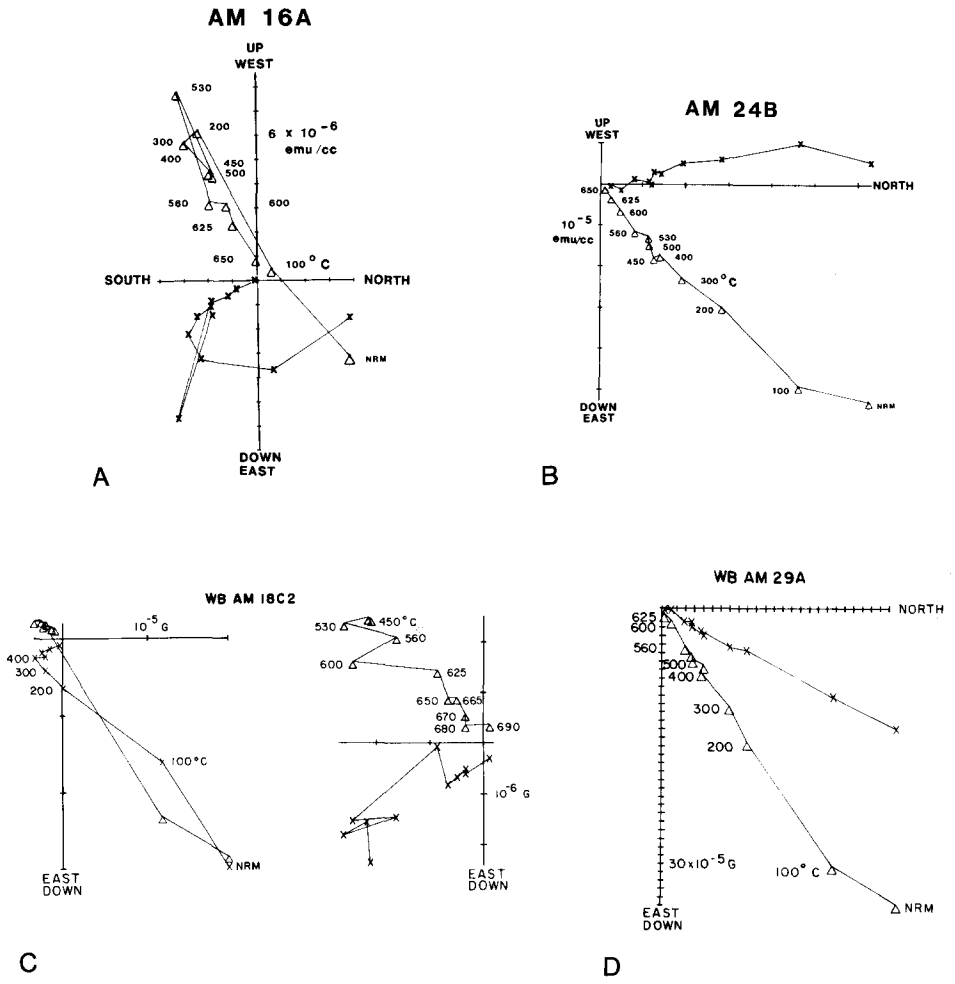


Fig.4. Vector demagnetization diagrams of samples from the Adobe Town Member, Washakie Formation, Washakie Basin. Plots (after Zijderveld, 1967) of magnetization vector behavior during progressive thermal demagnetization from NRM to 625°C. Normal polarity samples WB AM 24B (B), and WB AM 29A (D); and reverse polarity samples WB AM 16A (A), and WB AM 18C2 (C). X = horizontal component of vector, Δ= vertical component of vector, north is to right of page along horizontal axis, east and down is to bottom of page along vertical axis. Scale divisions are in 10^{-5} emu/cc (Gauss).

and coercivity spectra were observed and isolated in the pilot demagnetization studies. Therefore, all other samples were treated in a sequence of five to twelve demagnetization steps to isolate the component magnetizations.

Figure 4 shows representative plots of vector behavior during stepwise demagnetization of pilot samples. Samples 24B and 29A exhibit behavior characteristic of normally magnetized rocks in this section. A large secon-

dary component, in approximately the same direction as the primary magnetization, is removed by 300—400°C. This secondary magnetization is oriented in approximately the same direction as the geomagnetic field at this location and is probably caused by remagnetization in the present day field. Between 300—400°C and approximately 625—650°C the vector behavior becomes slightly more complex and less consistent, although a single magnetization component is indicated by the linear decay of the demagnetization (vector) curve to the origin. Above 650°C vector behavior is generally inconsistent and indicative of spurious or random magnetizations. In both samples the primary magnetization is oriented approximately to the north and down, as would be expected in a sample of normal polarity.

Samples 16A and $18C_2$ illustrate two types of magnetization behavior commonly observed in samples of reverse polarity. Both samples have a primary magnetization oriented generally towards the south (in these cases, to the SE) and up. In both samples the NRM vector was directed approximately to the N (or NE) and down, but subsequent vector behavior during demagnetization indicates that the NRM magnetization is dominated by a strong secondary component. Sample $18C_2$ exhibits the magnetization behavior typical of most reverse samples in this section; a very strong secondary magnetization is progressively removed at relatively low temperatures (by 300—450°C), then a much weaker, but more stable, primary magnetization is isolated at temperatures below 600—625°C. Samples from a number of sites behave similarly to sample 16A. In this sample a large secondary magnetization directed towards the north and down is rapidly removed at temperatures of 100—300°C. However, unlike sample $18C_2$ the primary magnetization initially isolated above 300°C is also of high intensity (although sample 16A shows a more marked intensity increase than most other similar samples). Following removal of the unstable component the vector plots of these samples exhibit generally linear decay to the origin, indicating isolation of a single, stable magnetization. Magnetizations above 625—650°C are very variable; they may be random, spurious, or stable. Stable magnetizations above 625°C may be of the same polarity but slightly different direction than, or of opposite polarity to, the primary magnetization.

Several samples, generally of low NRM intensity, show inconsistent or random motion of the vector demagnetization curve, yielding no useful information.

Stereonet plots of vector motion during demagnetization are shown for several of these pilot samples in Fig.3. Samples 24B (normal polarity) and $18C_2$ (reverse polarity) illustrate the generally greater directional stability of normal polarity samples, and the directional instability of vectors above about 625°C. In sample $18C_2$ the vector directions begin to cluster loosely (and presumably stabilize) at 400°C, and are relatively tightly clustered between 500 and 560°C. Above 600°C directions are more dispersed. The direction component of the magnetization in sample 24B appears moderately stable, but consistent vector motion continues throughout most of the

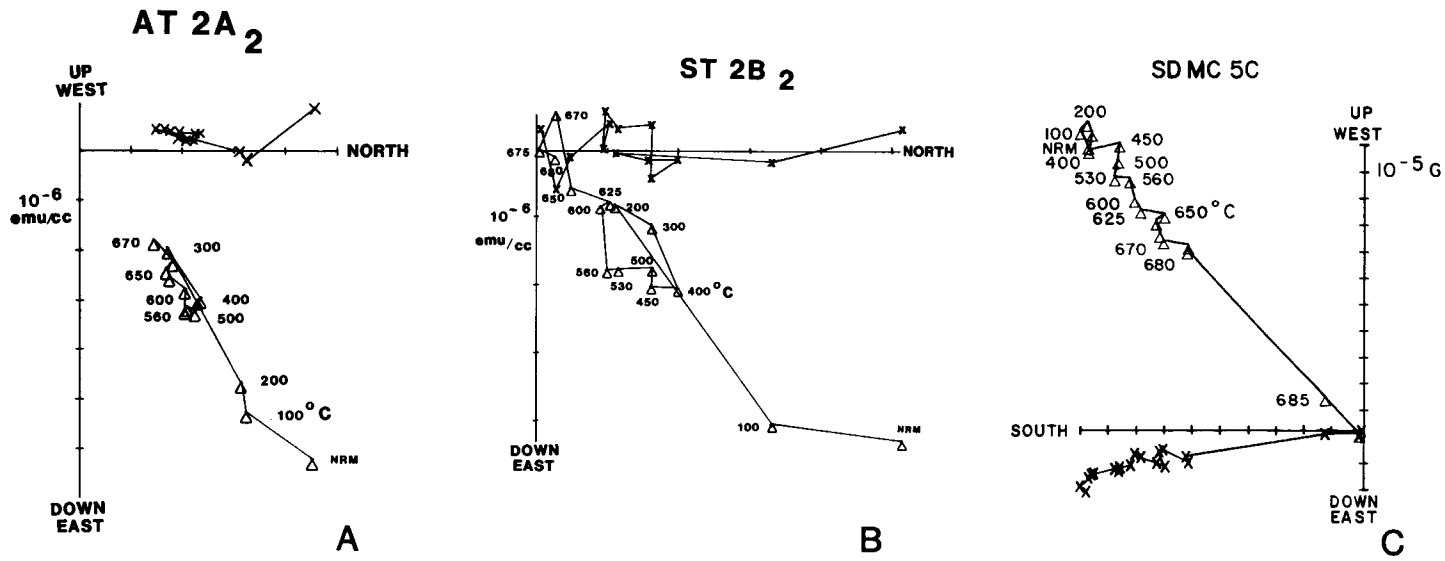


Fig.5. Vector demagnetization diagrams of samples from the San Diego Area. Plots (after Zijderveld, 1967) of magnetization vector behavior during progressive thermal demagnetization from NRM to 650°. Normal polarity samples AT $2A_2$ (A) from the Ardath Shale Type Section, and ST $2B_2$ (B) from the Scripps Formation Type Section; reverse polarity sample SD MC 5C (C) from the Murphy Canyon Section (Friars Formation). X = horizontal component of vector, Δ = vertical component of vector, north is to right of page along horizontal axis, east and down is to bottom of page, along vertical axis. Scale divisions are 10^{-6} emu/cc (Gauss) for A and B, and 10^{-5} emu/cc (Gauss) for C.

demagnetization, and the magnetization directions only cluster moderately well in the range of 400—560°C. Much of the directional change that occurs at higher temperatures is associated with a low intensity component. As in the Tepee Creek section, use of vector demagnetization diagrams allows a more precise determination of the orientation, stability and number of magnetization components.

San Diego area

Representative plots of vector behavior during progressive thermal demagnetization are shown in Fig.5. Samples ST $2B_2$ from the Scripps Formation Type Section and AT $2A_2$ from the Ardath Shale Type Section exhibit magnetization behavior commonly found in normal polarity rocks from the San Diego area. A large secondary component in sample ST $2B_2$ is removed during demagnetization to 200°C. Relatively steady decay of the magnetization oriented to the north and down occurs between 400 and 650°C. Sample AT $2A_2$ has a smaller secondary component that is removed by about 100—200°C. Again, relatively steady decay of a vector oriented to the north and down occurs between 400 or 500°C and 675°C. In both samples, an antiparallel, low blocking temperature magnetization component is removed between 200 and 400—500°C. The high blocking temperature component is isolated by 400—500°C, at relatively high remanent magnetization intensity (30—50% NRM intensity). Hematite clearly is a major contributor to the remanent magnetization in sample AT $2A_2$, as the magnetization intensity at 675°C is still 30% of the NRM intensity. Intensity of magnetization decreases sharply from 560 to 600°C (above the Curie temperature for magnetite), but a significant magnetic component remains above 600°C. The vector orientation of both the low and high blocking temperature components is about the same, indicating a stable, consistent direction of magnetization for both components. Many other normal polarity samples from the San Diego area sections exhibit a simpler magnetization.

The behavior of the magnetization vector for sample MC 5C (red color) is representative of some reverse polarity rocks in these sections. A small secondary component is removed by 200°C. Steady, straight line decay to the origin of a vector oriented to the south and up occurs from 200 to 685°C. Vector decay is consistent from 200 to 680°C, indicating a single, hematitic magnetization for this sample. There is a sharp intensity drop from 680—685°C (but little directional change), which indicates a narrow, high blocking temperature spectrum for the magnetization of this sample. It is likely that this magnetization is not due to fine-grained hematite pigment because of its very high blocking temperature (see Tauxe et al., 1980). Many other reverse polarity samples, particularly those that are not red in color, exhibit a more complex, multicomponent magnetization (with a wider blocking temperature spectrum).

Detailed stepwise thermal demagnetization studies to 675—690°C were performed on 19 samples (3 Delmar Formation, 3 Ardath Shale Type, 5

Scripps Formation Type, 3 Genessee Avenue, 3 Murphy Canyon, and 2 Mission Valley Formation sections) from the San Diego area sections. Detailed stepwise alternating field demagnetization studies were performed on 9 samples (2 Ardath Shale Type, 5 Scripps Formation Type, 2 Friars Formation Type Sections) from these sections. These studies reveal extremely variable behavior of the magnetization during demagnetization. Some samples have complex, multicomponent magnetizations with differing (but frequently overlapping) coercivity and blocking temperature spectra, others have relatively simple magnetizations. Because of the frequently complex behavior of the magnetization vector during demagnetization of the pilot samples, all samples were demagnetized either thermally, or with sequential alternating field and thermal demagnetization.

Magnetization components

Both vector demagnetization studies and very low median destructive treatments (MDT, Table II) indicate that a very low blocking temperature, frequently high intensity magnetization of normal polarity dominates the NRM of many samples. This magnetization generally is removed during demagnetization to 200—300°C and is interpreted as a secondary overprint acquired in the earth's present geomagnetic field. Stable magnetizations are preserved at intensities far below 50% (and even less than 10%) of the NRM values in most of these samples.

The mineralogy of the moderate blocking temperature (300—560°C) component is uncertain; it could be due to titanohematite [$x FeTiO_3.(1-x)Fe_2O_3$] or to titanomagnetite [$xFe_2TiO_4.(1-x)Fe_3O_4$]. Several Tepee Creek section and Washakie Formation pilot demagnetization samples have characteristic components with blocking temperatures below 600°C (e.g. sample TC10A, Fig.3). Some Washakie Formation pilot samples show a significant intensity decrease (but directional stability) upon demagnetization from 560 to 600°C (e.g. sample AM24B, Fig.5). Several San Diego area samples (e.g. ST $2B_2$, see discussion in previous section) show a major intensity drop, but directional stability, of the magnetization vector upon demagnetization from between 450°C and 560°C to the 600°C step. Other samples show a marked intensity increase at around 560°C, followed by an intensity decrease (but little directional change) at temperatures of 600°C and above. All of these behaviors indicate that primary titanomagnetite, with blocking temperatures less than 560—600°C, probably is the dominant carrier of the remanence. It is likely that the moderate blocking temperature component in many of the other samples with some high temperature hematitic remanence, also is caused by magnetite/titanomagnetite.

Further support for magnetite as the carrier of the primary remanence in the Wyoming sections is given by Shive et al. (1980). Shive et al. (1980, p. 945) report that petrologic and magnetic studies on sediments from the Aycross, Tepee Trail, and Wiggins Formations in the southeastern Absaroka

Range (just east of the East Fork Basin area) indicate that "The remanence of fresh samples is carried by magnetite, which is commonly in two size ranges — large grains in framework clasts, and finer grains in the matrix." It is extremely unlikely that magnetite/titanomagnetite is a secondary, diagenetic carrier of remanence.

High blocking temperature (greater than 600°C) hematite is a significant contributor to the remanence in many samples from all field areas. This is indicated by the presence of MDT's, and of a stable magnetization, at temperatures above the Curie temperature of magnetite (approximately 575°C). In most cases, this component has the same vector orientation as the higher intensity, moderate blocking temperature component (see above). In samples with two discrete magnetization components, if the first component is carried by magnetite (which is almost certainly primary), then the concordant directions between the moderate and high blocking temperature components in most of these samples indicates that the hematite also may be a primary remanence acquired at or shortly after deposition. However, the high and moderate blocking temperature components are of opposite polarity in several samples. It is unclear whether this hematite remanence is a primary DRM or a secondary diagenetic CRM. Only a few horizons have any reddish coloration that might indicate the obvious presence of a secondary hematite pigmentation. The presence of normal versus reversed polarity hematite is not correlated with lithology, sample color, grain size, etc., but is correlated in a coherent manner only with stratigraphic position. Discrete intervals of either polarity occur throughout the section; if the hematite remanence is secondary it would have been acquired several times, as it cannot be correlated in any simple or consistent manner with depth of burial, pressure, temperature, or other diagenetic processes. For instance, recent weathering and oxidation would be expected to form only normal polarity hematite oriented along the present geomagnetic field. The presence of both normal and reversed polarity hematite remanences supports the possibility that they are primary magnetizations acquired at the time of, or shortly after, deposition; they probably were not acquired through recent diagenetic processes.

Several samples in the San Diego sequence show a component with a relatively low blocking temperature spectrum (approximately 200—400°C; e.g. $ST2B_2$ and $AT2A_2$). This component is of uncertain origin and is oriented antiparallel to the moderate and high blocking temperature components discussed in the previous section. The blocking temperature (about 400°C) for this component is probably too high for a viscous magnetization or secondary magnetization carried by goethite, but seems reasonable for a very fine-grained, secondary hematite pigmentation or staining or for a viscous magnetization carried by hematite. However, both sample ST $2B_2$ and AT $2A_2$ are yellowish-brown or yellowish-tan. This magnetization behavior is uncommon in San Diego area samples.

Susceptibility

Trends of susceptibility behavior during progressive thermal demagnetization can be used to identify regions in which physico-chemical or mineralogical changes may be occuring in a sample. Samples from the Tepee Creek and Aycross Formation sections (see Flynn, 1983) generally exhibited very small susceptibility increases or steady susceptibility decreases upon heating between 100 and 625°C or 650°C. All of these samples show a sharp, large decrease in susceptibility at 650—660°C followed by a steady susceptibility decrease to 680—690°C.

Kent and Opdyke (1978) and Dunlop (1972) attributed major susceptibility increases during progressive thermal demagnetization to major mineralogical changes, such as formation of large amounts of magnetite from non-magnetic ferric oxide precursors or hematite. The cause of the small susceptibility decreases in the East Fork Basin samples, and of the major susceptibility decreases above 650°C in all the Tepee Creek samples, is uncertain. Alteration or oxidation of high susceptibility minerals, such as magnetite, to lower susceptibility minerals, such as hematite, in an open air system may be one possible explanation for the observed susceptibility decreases. These studies do not indicate any significant alteration of magnetic mineralogy, or magnetic remanence properties, during progressive thermal demagnetization below 650°C.

NRM magnetization/susceptibility (J_{nrm}/x)

Magnetic susceptibility measurements were made on samples from all three field areas. Measurements were made on all samples prior to thermal demagnetization, and following any subsequent thermal demagnetization steps. Ranges of susceptibilities and J_{nrm}/x values for samples from these sections are given in Table II.

Irving (1964, p. 92), Evans and McElhinny (1969; citing Stacey, 1967), and McElhinny (1973, p. 58) suggested that in igneous rocks J_{nrm}/x ratios greater than 0.5—1.0 are associated with stable NRM's probably carried by magnetite, and values less than 0.1 usually indicate significant unstable magnetic components. It is likely that stable magnetizations in detrital sediments (DRM) are present at J_{nrm}/x values significantly less than those for igneous rocks, because detrital sediments generally are less efficient recorders of an applied magnetic field. For a given applied field J_{nrm} is lower in sediments than in igneous rocks (consistently lower average H_c), and therefore for any given bulk susceptibility the J_{nrm}/x ratio should consistently be lower in sediments.

In this study, stable magnetizations (determined by vector demagnetization) are exhibited by most samples with J_{nrm}/x ratios of between 0.1 and 5.0. Many samples with ratios greater than 5.0 also have stable magnetizations. Magnetic susceptibility data support the presence of a stable remanence,

probably carried by magnetite, in most sections. There is no correlation between magnetization vector stability and J_{nrm}/x variation, lithology, color, or other physical characteristics in these samples.

MAGNETIC POLARITY STRATIGRAPHY

Reliability of data

In this section I present composite magnetic polarity stratigraphies for the East Fork Basin area, Washakie Basin, and San Diego area strata. Primary vector directions were used to determine mean directions of magnetization (Fisher, 1953) for each site in the individual sections sampled in each area. I then use these site mean directions (and VGP's determined from them) to infer the polarity of the earth's magnetic field at the time of acquisition of the primary remanent magnetization for each sampling horizon. The stratigraphic sequence of site polarities is used to construct a composite magnetic polarity stratigraphy for the area.

In order to construct an accurate magnetostratigraphy it is essential to document that the acquisition of the primary magnetization was contemporaneous with the deposition of the sampled unit. The demagnetization procedures discussed above generally were successful in isolating a stable magnetization for the sediments from all three field areas. The magnetization vectors in the San Diego area sequence are frequently less stable, and more difficult to isolate, than in the East Fork Basin area and Washakie Basin sequences.

Two field tests for magnetization stability (fold test and reversals test, see McElhinny, 1973) can be applied to the data from the East Fork Basin area, the Washakie Formation, and the San Diego area. A fold test for magnetic stability is not possible for the San Diego area samples because most of the sampled strata are horizontal, or dip only very gently (approximately 2°, generally to the east). Only sites with statistically significant, non-random mean resultant vectors are considered in these stability tests.

Mean magnetization directions for the East Fork Basin area and Washakie Formation, and San Diego area sections are given in Tables III—V. Reversals tests were applied to the Class I *bedding corrected* data from the Aycross Formation (East Fork Basin area, 13 positive inclination sites, 12 negative inclination sites), Tepee Creek section (East Fork Basin area, 4 positive, 14 negative), composite East Fork Basin area (17 positive, 26 negative), Kinney Rim Member section (Washakie Formation, 3 positive, 12 negative), Skull Creek Section (Washakie Formation, 6 positive, 4 negative), Adobe Town Member section (Washakie Formation, 6 positive, 11 negative), composite Washakie Formation (15 positive, 27 negative), and composite San Diego area sections (25 positive, 7 negative). For each of these sections, except the Adobe Town Member (see below), inversion of the negative inclination site means (and paleopoles) to the same hemisphere as the positive inclination

TABLE III

Paleomagnetic results from the East Fork Basin area, Wyoming sections

	Decl.	Incl.	Paleopole		N	K	C of C	R
			Lat.	Long.				
Aycross Formation section bedding corrected mean, positive inclination sites	356.0	59.8	85.9	118.3	13	14.8	11.1	12.1914
Aycross Formation section bedding corrected mean, negative inclination sites	167.7	—49.8	—73.8	291.5	12	19.7	10.0	11.4430
Aycross Formation section no bedding correction mean, positive inclination sites	358.2	53.6	80.5	79.5	13	13.8	11.5	12.1347
Aycross Formation section no bedding correction mean, negative inclination sites	168.5	—44.1	—70.0	282.1	12	20.0	9.9	11.4511
Aycross Formation section bedding corrected mean, all sites	351.4	55.0	79.7	112.8	25	16.2	7.4	23.5242
Aycross Formation section no bedding correction mean, all sites	352.9	49.1	75.4	95.2	25	15.6	7.5	23.4683
Tepee Creek section bedding corrected mean, positive inclination sites	17.0	72.3	72.4	281.7	4	31.9	16.4	3.9062
Tepee Creek section bedding corrected mean, negative inclination sites	166.3	—56.2	—77.5	312.1	14	14.5	10.7	13.1093
Tepee Creek section no bedding correction mean, positive inclination sites	14.2	73.9	71.4	273.0	4	32.1	16.4	3.9066
Tepee Creek section no bedding correction mean, negative inclination sites	164.1	—57.0	—76.6	319.5	14	14.6	10.7	13.1101
Tepee Creek section bedding corrected mean, all sites	350.4	60.3	82.6	145.8	18	14.4	9.4	16.8254
Tepee Creek section no bedding correction mean, all sites	348.0	61.3	81.1	156.8	18	14.4	9.4	16.8259
Composite — Aycross and Tepee Creek bedding corrected mean, positive inclination sites	359.3	63.0	88.0	228.1	17	15.9	9.2	15.9967
Composite — Aycross and Tepee Creek bedding corrected mean, negative inclination sites	167.0	—53.2	—76.0	300.9	26	16.8	7.1	24.5127
Composite — Aycross and Tepee Creek bedding corrected mean, all sites	351.1	57.2	81.2	123.6	43	15.5	5.6	40.3067
Composite — Aycross and Tepee Creek no bedding correction mean, all sites	351.2	54.2	78.9	111.0	43	14.3	5.9	40.0656

TABLE IV

Paleomagnetic results from the Washakie Basin, Wyoming sections

	Decl.	Incl.	Paleopole		N	K	C of C	R
			Lat.	Long.				
Kinney Rim Member section bedding corrected mean, positive inclination sites	26.3	60.1	70.2	332.6	3	29.4	23.1	2.9320
Kinney Rim Member section bedding corrected mean, negative inclination sites	162.7	−67.6	−74.7	25.6	12	7.6	16.7	10.5609
Kinney Rim Member section no bedding correction mean, positive inclination sites	346.6	68.1	76.3	213.8	3	29.3	23.1	2.9318
Kinney Rim Member section no bedding correction mean, negative inclination sites	123.9	−59.7	−48.3	0.0	12	7.6	16.8	10.5566
Skull Creek section bedding corrected mean, positive inclination sites	337.1	52.6	70.2	145.4	6	16.9	16.7	5.7043
Skull Creek section bedding corrected mean, negative inclination sites	166.1	−51.7	−75.8	307.7	4	49.1	13.2	3.9390
Skull Creek section no bedding correction mean, positive inclination sites	331.2	58.5	68.0	166.1	6	17.2	16.5	5.7107
Skull Creek section no bedding correction mean, negative inclination sites	161.7	−57.9	−75.7	337.5	4	48.6	13.3	3.9383
Adobe Town Member section bedding corrected mean, positive inclination sites	19.6	56.8	74.4	348.5	6	50.9	9.4	5.9018
Adobe Town Member section bedding corrected mean, negative inclination sites	136.9	−59.1	−57.5	354.0	11	11.2	14.2	10.1079
Adobe Town Member section no bedding correction mean, positive inclination sites	6.9	68.1	78.7	274.4	6	50.0	9.5	5.9002
Adobe Town Member section no bedding correction mean, negative inclination sites	116.1	−57.3	−41.7	359.8	11	11.0	14.3	10.0965
Composite — all Washakie sections bedding corrected mean, positive inclination sites	2.9	57.9	86.6	28.8	15	17.3	9.4	14.1927
Composite — all Washakie sections bedding corrected mean, negative inclination sites	151.9	−62.2	−69.2	357.7	27	9.7	9.3	24.3306
Composite — all Washakie sections no bedding correction mean, positive inclination sites	346.3	65.2	78.4	198.2	15	22.9	8.1	14.3900
Composite — all Washakie sections no bedding correction mean, negative inclination sites	126.4	−59.4	−49.9	358.8	27	9.7	9.3	24.3417
Composite — all Washakie sections bedding corrected mean, all sites	344.2	61.5	78.2	174.4	42	10.7	7.0	38.1799
Composite — all Washakie sections no bedding correction mean, all sites	319.2	62.8	60.1	181.3	42	10.8	7.0	38.2343

TABLE V

Paleomagnetic results from the San Diego area, California sections

	Decl.	Incl.	Paleopole		N	K	C of C	R
			Lat.	Long.				
San Diego Area sections, composite mean, positive inclination sites	356.2	52.7	86.8	163.3	25	11.5	8.9	22.9181
San Diego Area sections, composite mean, negative inclination sites	176.2	−37.5	−77.8	259.8	7	18.5	14.4	6.6754
San Diego Area sections, composite mean, all sites	356.1	49.3	85.8	116.0	32	12.0	7.7	29.4105

site means indicates that the means (and paleopoles) for each section are indistinguishable at the 95% confidence level. This antiparallel relationship between the reversed and normal polarity samples is a positive reversals test. A positive reversals test supports isolation of a stable primary magnetization, of both reversed and normal polarity, during demagnetization of these samples.

For the Adobe Town Member section, the positive and inverted negative inclination means (and paleopoles) are distinguishable at the 95% confidence level. Both the normal polarity sample inclination (+56.8°) and reversed polarity sample inclination (−59.1°) are close to the expected site inclination of ±60.4° for a locality at the latitude of the Washakie Basin. However, the declinations of both the normal (19.6°) and reversed (136.9°) polarity samples deviate from the expected values of 0° and 180°, respectively. The particularly large declination deviance in the reversed polarity sample is probably due to a small secondary overprint that has not been removed completely during demagnetization. The behavior of the Adobe Town Member samples during demagnetization reflects the presence of this secondary component. The decay of the magnetization vectors (on vector demagnetization plots) is less consistent and less stable, and the clustering of site mean vectors (reflected in k and R values for each site) is generally poorer in the Adobe Town Member section than in the Kinney Rim Member and Skull Creek sections.

Upon inversion for composite site means from all three Washakie Formation sections, the negative and positive inclination site means are indistinguishable at the 95% confidence level, but the paleopoles to these means are barely distinguishable at the 95% confidence level. The indication of two possibly separate populations for the paleopoles probably is due to the influence of the small secondary component in the Adobe Town Member samples.

Fold tests on both the East Fork Basin area and Washakie Formation samples yield inconclusive results. The precision parameter (k) values are so similar for unfolded, bedding corrected samples and in situ, uncorrected site means, that samples from these sequences neither fail nor pass the fold test unambiguously (see Tables III and IV). Acquisition of remanence prior to folding cannot be documented, but neither can one conclude that there was acquisition of a secondary remanence after folding. The ambiguous fold test,

and closely similar k values before and after application of bedding corrections may be due to: (1) shallow bedding dips and relatively narrow range of bedding strikes for these tilted strata, or (2) "unfolding" of strata about a single horizontal axis (most common paleomagnetic procedure), when unfolding about a series of rotation axes may reflect more appropriately the true tectonic tilting history (see MacDonald, 1980).

The mean composite direction for all Class I site means (negative inclination sites inverted; all sites bedding corrected) from the East Fork Basin area, Washakie Formation, and San Diego area are given in Tables III—V. The paleopole to the composite mean magnetization direction for the East Fork Basin area is latitude = 81.2°N, longitude = 123.6°E, for the Washakie Formation is latitude = 78.2°N, longitude = 174.4°E, and for the San Diego area is latitude = 85.8°N, longitude = 116.0°E. .

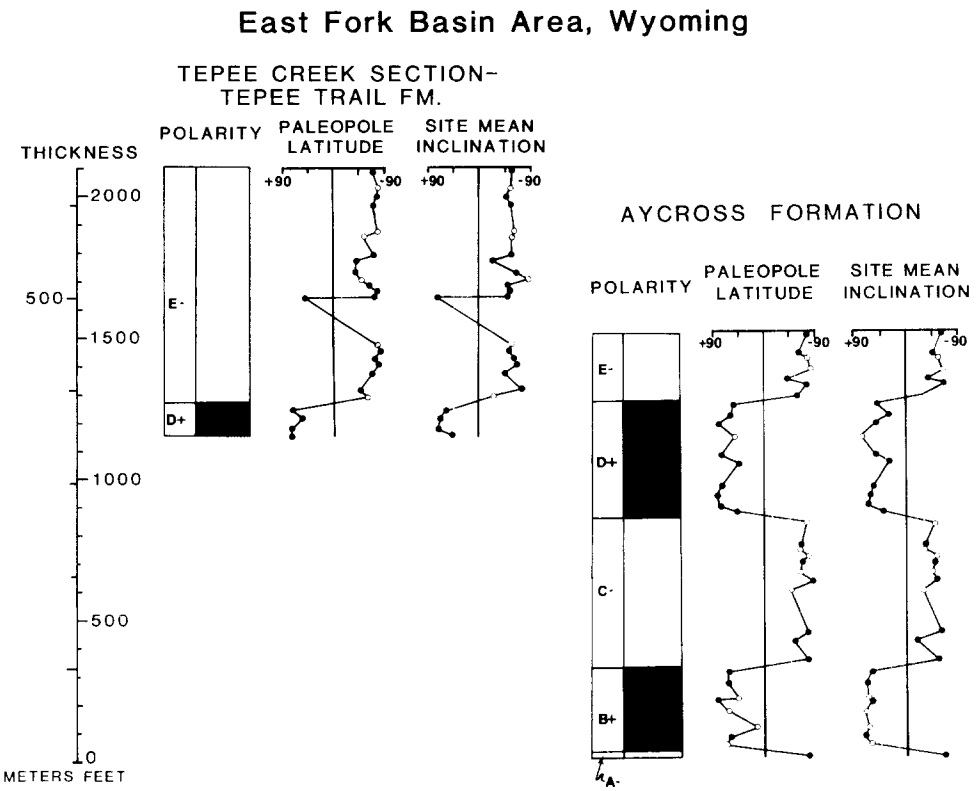


Fig.6. East Fork Basin area, Wyoming magnetostratigraphy. Tepee Creek section (Tepee Trail Formation) and Aycross Formation section magnetic polarity data plotted versus stratigraphic thickness. Both site mean inclination and paleopole latitude (determined from site mean directions) are given for each section. Closed circles are Class I reliability sites (see text), open circles are Class II—IV reliability sites (see text). Magnetic polarity sequences are plotted to the left of the paleomagnetic data columns. Polarity intervals are lettered from A to E based on superposition; + or — refers to intervals of predominantly normal or reverse polarity, respectively. The two sections are aligned using the top of polarity interval D+ as a datum.

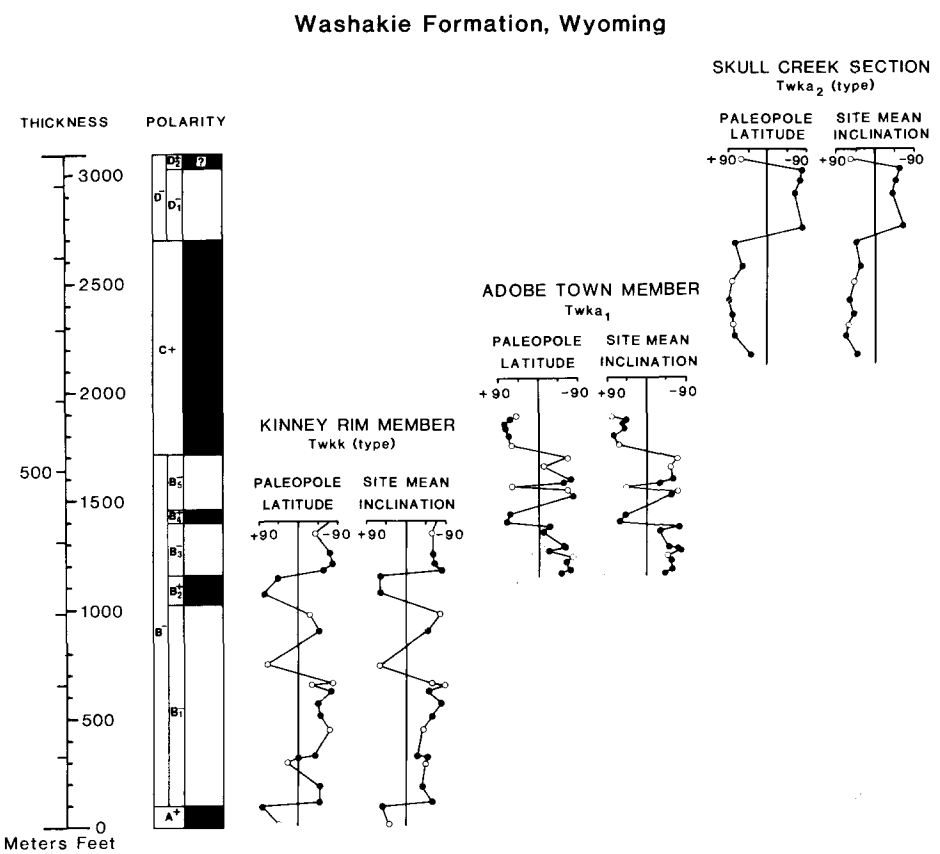


Fig.7. Washakie Basin, Wyoming magnetostratigraphy. Kinney Rim Member, Adobe Town Member and Skull Creek sections (all Washakie Formation) magnetic polarity data plotted versus stratigraphic thickness. Figure constructed as discussed for Fig.6. Numerical subscripts (e.g. B_1—) refer to subdivisions of the major polarity intervals. The three sections are aligned by use of marker beds that have been physically, laterally traced from section to section to produce the composite magnetostratigraphic section.

Magnetic polarity stratigraphy

Figures 6—8 show the magnetic polarity sequences for the East Fork Basin area, Washakie Formation, and San Diego area. The reliability classification of Opdyke et al. (1977) was used to evaluate the reliability of site magnetization directions for polarity determinations. Closed circles are Class I reliability sites, which Opdyke et al. (1977) define as any site that is statistically significant (using Watson's, 1956a, b statistical tests). Open circles are Class II—IV reliability sites. Opdyke et al. (1977) define Class II sites as having only two samples available per site, both yielding concordant directions; Class III sites as those possessing statistically random magnetization directions, but two of three samples give similar polarity information; and Class IV

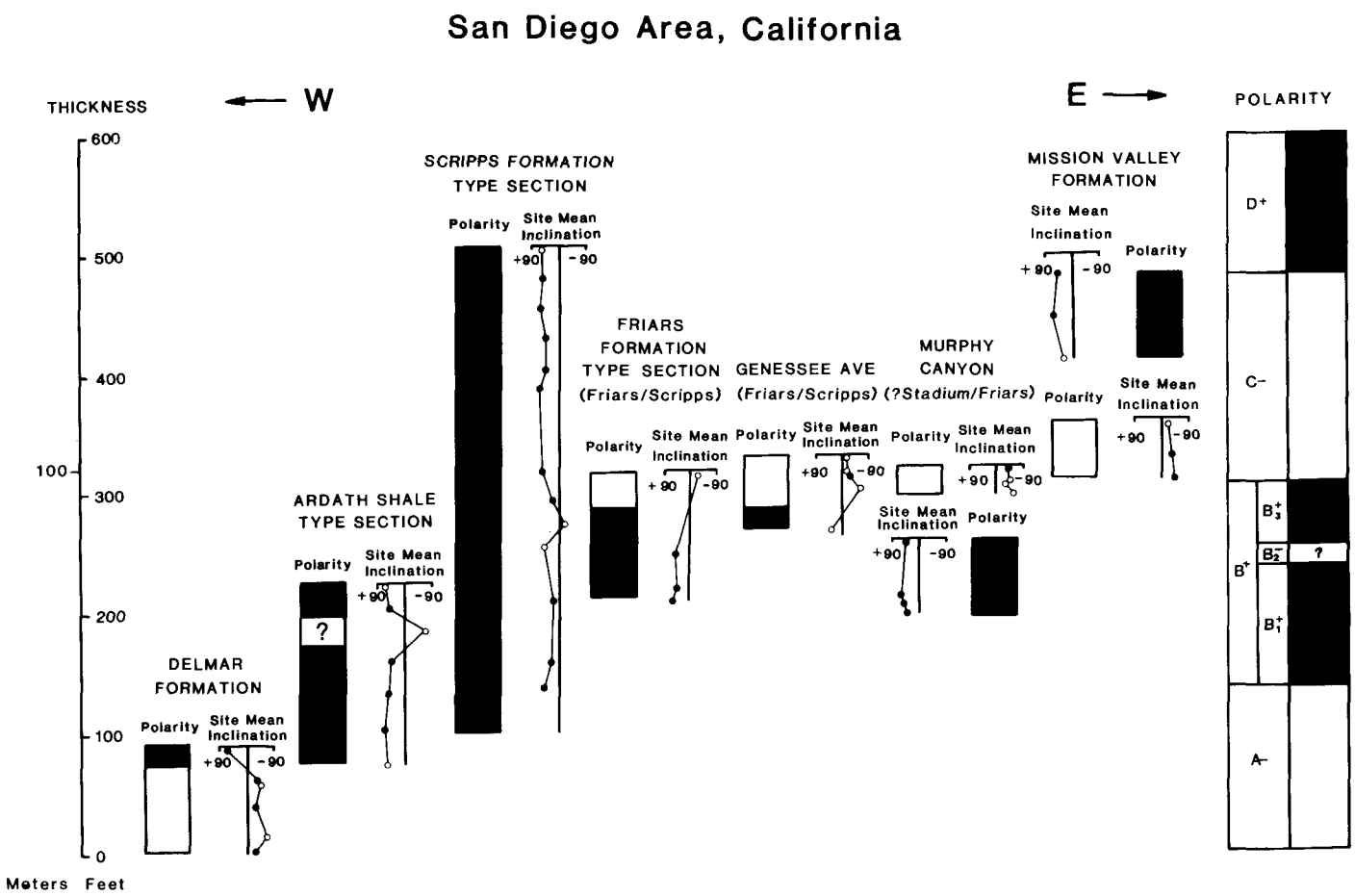


Fig.8. San Diego area, California magnetostratigraphy. Delmar Formation, Ardath Shale Type, Scripps Formation Type, Friars Formation Type, Genessee Avenue, Murphy Canyon, and Mission Valley Formation Sections, magnetic polarity data plotted versus stratigraphic thickness. Figure constructed as discussed for Fig.6. The E (east) and W (west) plotted at the top of the figure indicate the *general* W—E, marine to non-marine, older to younger trend of the individual sections.

sites as those in which a strung distribution is evident, but the directions are statistically random. Both Class I and Class II—IV reliability sites yield significant, complementary information about the magnetic polarity of the strata in all three sequences. Class V sites of Opdyke et al. (1977) have random magnetization vectors, no consistent trends in vector directions, and circles of confidence (alpha 95) greater than 90°; these sites are not used for polarity determinations.

I follow Tauxe (ms.) in using Class I sites, for which specimens have been thermally demagnetized, to determine normal polarity intervals. I also use specimens from Class II—IV sites, that have been thermally demagnetized, to help define these normal polarity zones. Thermal demagnetization is essential to demonstrate that normal polarity remanences are primary rather than secondary. Only polarity intervals represented by more than one site are recognized in this study, except at the base or top of a sequence where single sites may be used to tentatively define a new polarity interval (e.g. A— on Fig.6). This removes the influence of single, possibly anomalous sites within intervals of one predominant polarity, but allows recognition of potential polarity reversals at the very base or top of a section.

East Fork Basin area

Five magnetic polarity intervals, A— to E— (see Fig.6) are present in this sequence. Within this sequence there are 26 reverse polarity and 18 normal polarity Class I sites, 13 reverse polarity and 5 normal polarity Class II—IV sites, and 8 Class V sites. The validity of the single normal polarity site within the reversed interval E— of the Tepee Creek section is suspect because it was sampled (in an attempt to gain polarity information in a barren interval) in the medium-grained sandstone *matrix* of a conglomeratic interval.

I use the top of normal polarity interval D+ as a datum to align the Tepee Creek and Aycross Formation sections. These two sections cannot be directly, physically correlated because of the lack of direct stratigraphic continuity between the two sections (separated by the buried crest of the Washakie Range, and individual beds are not very continuous laterally). Several lines of evidence support the use of the top of this normal polarity interval as an equivalent horizon in both sections. Faunal evidence indicates that Bridgerian mammals occur in the top of the Aycross Formation section (and possibly low in the Tepee Trail Formation near the Tepee Creek section) and earliest Uintan mammals occur within a horizon near the middle of the Tepee Creek section. Bridgerian is temporally very short (see below) and it is unlikely that there would be a significant time gap between the top of the Aycross Formation and the middle of the Tepee Creek sections (such that the D+ polarity interval in the two areas would represent two different polarity zones, rather than one). Synchronous correlation of the top of polarity interval D+ between the two sections seems to be the most reasonable interpretation of the available data.

Corroboration of the magnetic polarity sequence proposed in Fig.6 is

found in the work of Shive et al. (1980), conducted nearby to the east of the East Fork Basin area. Shive et al. (1980) state that the Tepee Trail Formation in their study area contains a normal to reverse polarity transition about 270 m above the base of the 760 m thick section. Reconnaissance sampling of the underlying Aycross Formation indicated a possible normal polarity for that unit. This pattern agrees closely with the magnetic polarity sequence developed for the East Fork Basin area, and supports recognition of only a single normal polarity interval near the top of the Aycross Formation and at the base of the Tepee Trail Formation.

Washakie Formation

Four major magnetic polarity intervals, A+ to D— (see Fig.7) are present in this sequence. Within this sequence there are 26 reverse polarity and 15 normal polarity Class I reliability sites, 10 reverse polarity and 8 normal polarity Class II—IV sites, and 13 Class V sites. Two single, normal polarity Class II—IV sites occur within reversed polarity intervals within this sequence (Adobe Town Member section, at about the 466 m, or 1530 ft, level in the composite section; Kinney Rim Member section, at about the 224 m, or 738 ft, level in the composite section). These points are anomalous, and probably reflect a spurious or secondary magnetization. These single points are not used to define polarity intervals (see above). At a stratigraphic level of about 91.5 m (300 ft) in the Kinney Rim Member section there are two paleopole positions that might suggest a normal polarity interval within polarity zone B_1-. However, since only one (Class II—IV) site clearly has a normal polarity paleopole latitude, and because these two sites both have negative inclinations, I do not consider them representative of a distinct, valid normal polarity interval. The anomalous paleopole latitudes for these two sites are due to their anomalous site declinations.

The three sections in this composite sequence are correlated and aligned by the use of shared marker beds that can be physically traced laterally from section to section. There is stratigraphic overlap between the tops and bottoms of each of the sections, although magnetic data have not been obtained for all the intervals of overlap. The magnetic polarity patterns in the intervals of stratigraphic overlap are concordant between the sections (e.g. the top of the Kinney Rim Member and bottom of the Adobe Town Member sections, Fig.7). This concordance and repeatability supports the inferred pattern of magnetic polarity stratigraphy and the evidence cited above for the isolation of a stable, primary remanence in these sediments.

The top of polarity interval B_2+ marks the base of the lower brown sandstones and the base of the Adobe Town Member of the Washakie Formation. This upper boundary therefore is an unconformity (see discussion above), and both polarity interval B_2+ and B_3- may be relatively shorter than the lengths of the polarity intervals they represent, because of sediment not present because of erosion or non-deposition. However, the presence of Bridgerian mammals both below and above this unconformity, and the short

temporal duration of Bridgerian time (see below), suggest that this erosional/non-depositional hiatus probably did not represent a very long time interval.

San Diego area

Four major magnetic polarity intervals, A— to D+ (see Fig.8) are present in this sequence. Within the San Diego sequence there are 7 reverse polarity and 24 normal polarity Class I reliability sites, 12 reverse polarity and 6 normal polarity Class II—IV sites, and 6 Class V sites. The negative inclination site near the middle of the Scripps Formation type section is considered an anomalous data point. The behavior of the magnetization vectors for the samples from this site was not very stable, and the resultant is of very low inclination.

The sections are aligned in Fig.8 using the known stratigraphic relationships between the formations in the San Diego area (see Kennedy and Moore, 1971; Kennedy, 1975). Direct stratigraphic continuity between the sections in this area is difficult, or impossible, to document because of urban development, vegetation, faulting, etc. Further, lateral variation in the age and development of various formations may result in laterally transgressive ages for some formations. Therefore, some of the stratigraphic relationships are hypothetical and inferential. However, the general stratigraphic relationships portrayed in Figs.1 and 8 are likely to be approximately correct. The magnetostratigraphic patterns developed for these sections are consistent with, and support the validity of, the inferred stratigraphic relationships (e.g. there are no major conflicts between the polarities and the inferred age and stratigraphic relationships between and within formations). In general the section shown on Fig.8 trends from west to east, and the sections sampled become progressively younger and more non-marine from west to east.

The stratigraphic expansion of the Scripps Formation Type Section normal polarity interval may be due to the fact that this part of the formation was deposited in close proximity to, or as part of, a major submarine channel (see Lohmar and Warme, 1979; Lohmar et al., 1979). The sedimentation rate in such an environment might be expected to be very high.

Several sections presented possible correlation difficulties. The relative stratigraphic position of the polarity segments in the Murphy Canyon section and Mission Valley Formation section is uncertain. The stratigraphic relationships shown for the two polarity segments in each section are probably correct, but the lack of direct continuity between the sections makes this correlation less certain. The two Murphy Canyon sections are in close physical proximity, and the proposed superpositional relationships are probably correct. However, the two Mission Valley Formation segments are separated by a significant areal distance. I have inferred the relative stratigraphic position of these two segments using their distance from the contact with the overlying Pomerado Conglomerate, and the inferred relative age transgression (based on E—W geographic location, see Kennedy, 1975) for the Mission Valley Formation sediments.

Detailed correlation of the local magnetic polarity zonations in each field area to the standard magnetic polarity time scale is discussed below.

RECOGNITION OF A NEW TIME INTERVAL — THE SHOSHONIAN (EARLIEST UINTAN) LAND MAMMAL SUBAGE

Previous discussions indicate that Bridgerian and/or Uintan mammal faunas occur in all three of the field areas of this study. Further, all three areas contain strata of "earliest Uintan" age, while the Washakie Basin and East Fork Basin area sections definitely, and the San Diego area sections probably, span the Bridgerian/Uintan boundary. "Earliest Uintan" faunas have been recognized in these areas because several assemblages appear to be intermediate in age between previously known Bridgerian and Uintan faunas. This intermediacy is recognized through the stratigraphic and temporal overlap of taxa that elsewhere have disjunct temporal ranges within either the Bridgerian or Uintan, and through the occurrence of taxa that are morphologically more advanced than related taxa from known Bridgerian faunas or that are more primitive than related taxa from known Uintan faunas.

In San Diego, taxa with Uintan first appearances (West et al., in press — *Amynodon*, *Simidectes*, *Protoreodon*, and *Leptoreodon*) or that are characteristic of, or only found in, the Uintan (*Protoreodon*, *Amynodon*, *Protylopus*, *Ischyrotomus*, *Leptoreodon* sp. cf. *L. marshi*, and *?Macrotarsius* sp. near *M. jepseni*) co-occur with Bridgerian index taxa (*Notharctus* sp. near *N. robustior* and *Microsyops* sp. cf. *M. annectens*) or taxa that are found only in, or are characteristic of, the Bridgerian (*Hemiacodon* sp. near *H. gracilis*, *Washakius woodringi* [near *W. insignis*], *Microsyops* sp. cf. *M. annectens*, *Notharctus*, *Omomys*, *Uintasorex*, *Sciuravus*, *Herpetotherium* (=*Peratherium*), and *Apatemys*).

Similarly, in Bone Bed A of the Tepee Trail Formation type section, taxa with Uintan first appearances (*Oligoryctes*, *Amynodon*, eomyid rodents, and selenodont artiodactyls) and Uintan index taxa (*Amynodon advenus*, *Achaenodon*, and possibly *Epihippus*) co-occur with characteristic Bridgerian taxa (*Apatemys*, *Hyopsodus*, *Uintasorex*, *Herpetotherium* [=*Peratherium*], and *Sciuravus*) and taxa that are elsewhere known only from the Bridgerian (*Uintasorex parvulus*, *Trogolemur* cf. *T. myodes*, and cf. *Tetrapassalus* sp.).

The fauna from the upper part of the lower Adobe Town Member, Washakie Formation (bracketed by Bridgerian and Uintan faunas) is of uncertain temporal affinity; it contains characteristic Bridgerian taxa (*Paramys*, *Sciuravus*, *Hyopsodus* and *Miacis*), taxa with Uintan last occurrences (*Paramys*, *Sciuravus*, *Hyopsodus* and *Uintatherium*), and a Bridgerian index taxon (*Notharctus robustior* — which also occurs in San Diego, coeval with Uintan taxa).

What is the resolution of this apparent anomaly in temporal distribution of taxa in these areas? Given the apparent intermediate nature of several of these faunas, is it possible to define rigorously and recognize a discrete

chronostratigraphic or chronologic interval that lies temporally between the previously (Wood et al., 1941; West et al., in press) defined Bridgerian and Uintan mammal ages? I propose an answer to these questions below.

Based on the information presented previously, I believe it is possible to define, and recognize, a temporal unit between the classic Bridgerian and Uintan Ages. This unit may be *defined* by the first occurrence of the amynodontoid rhinoceros *Amynodon*. It is characterized, and can be recognized, by the occurrences of *Amynodon*, *Ischyrotomus*, *Metarhinus*, *Protylopus*, *Apatemys*, *Herpetotherium* (=*Peratherium*), *Leptoreodon*, *Protoreodon*, *Dilophodon*, *Epihippus*, *Sciuravus*, *Microsyops*, *Uintasorex*, *Nyctitherium*, *Notharctus robustior*, *Achaenodon*, *Trogolemur myodes*, and *Hyopsodus paulus* and *Hyopsodus* n.sp. (East Fork Basin; Flynn, m.s.). First appearances include *Amynodon*, selenodont artiodactyls (*Leptoreodon*, *Protoreodon*, *Protylopus*), *Epihippus*, and *Achaenodon*. Last occurrences include *Uintasorex*, *Northarctus robustior*, *Trogolemur myodes*, *Microsyops annectens*, and *Hemiacodon*. No genera are presently known to be restricted to this temporal interval. Diagnostic faunas of this age are found in the type section and type area of the Tepee Trail Formation, Fremont County, Wyoming, and in the Friars and Mission Valley Formations, San Diego, California. A fauna of this age should be found in the middle to upper part of the lower Adobe Town Member, Washakie Formation, Wyoming.

Should this temporal unit be considered part of the Bridgerian or Uintan, or should a new unit be erected? The Bridgerian was defined (Wood et al., 1941) as the temporal duration of the strata and fauna of Matthew's (1909) Bridger A—D. West et al. (in press) recommend that the definition of Bridgerian be expanded to include Bridger E, which West and Hutchison (1981) have shown contains a standard Bridgerian fauna. Accepting either of these two definitions, it is impossible to include this new temporal unit within the Bridgerian. Faunally, and temporally, it is quite distinct from the previously defined Bridgerian. However, the "unfortunate inclusion of the unfossiliferous Uinta A in the Uintan definition (by Wood et al., 1941), thus preventing faunal characterization of the early part of the Uintan" lamented by West et al. (in press) may allow inclusion of this new temporal unit within the previous (but expanded) definition of the Uintan. There is an obvious temporal gap between the known Bridgerian and Uintan, represented by this new temporal unit; this temporal unit cannot be encompassed within the definition of the Bridgerian; however, all, or part, of the unfossiliferous Uinta A must be temporally equivalent to all, or part, of this time unit because it is overlain by strata of early (Uinta B) and late (Uinta C) Uintan age. It seems possible, then, to consider this new temporal unit as part of the previously defined Uintan (equivalent to all, or part, of the poorly characterized Uinta A time). If we choose to consider this unit as part of the Uintan, the most appropriate terminology for this subdivision might be "earliest Uintan", or possibly "early Uintan" if "Uinta B time" is considered "medial Uintan" instead of its common assignment as "early Uintan". At

present, it seems preferable to consider this new temporal unit as earliest Uintan in age (= more than, all of, or part of Uinta A time) rather than early Uintan, so as to avoid confusion with the more common use of early Uintan to refer to Uinta B (+ Uinta A) time. "Uinta C time" would still be considered late Uintan.

The other terminological alternative for this new unit is to propose a new name, the Shoshonian, for this interval of time. Such an approach is both appropriate, and reasonable, and it circumvents many of the complex definitional and nomenclatural problems discussed above. I propose Shoshonian because of the presence of extensive faunas of Earliest Uintan age in close proximity to the Shoshone National Forest and Shoshone Indian Reservation. Given the possibility of readily including this time unit within the Uintan, I presently prefer to consider this unit as the earliest interval of the Uintan Land Mammal Age — the Shoshonian Land Mammal Subage.

BIOCHRONOLOGIC CORRELATION OF THE THREE FIELD AREAS

A rigorous analysis and biostratigraphic typification of this new time interval in an area with a complete sequence of Bridgerian through Uintan faunas, ultimately will permit proper stratotypification of a Shoshonian Substage upon which erection of a formal subage must be based (see Tedford, 1970; Flynn et al., 1984).

There are several taxonomic occurrences in common that support the precise correlation of the Shoshonian (Earliest Uintan) portions of the sequences from the East Fork Basin and San Diego areas. *Nyctitherium*, *Uintasorex*, *Microparamys*, *Sciuravus*, *Dilophodon*, *Herpetotherium* (= *Peratherium*), *Apatemys*, *Reithroparamys*, and *Amynodon* all occur in both the San Diego and East Fork Basin area Shoshonian (Earliest Uintan) assemblages. Elsewhere (see West et al., in press) *Uintasorex* and *Dilophodon minusculus* are found only in the Bridgerian (*D. leotanus* only in the Uintan), *Amynodon* is known only from the Uintan, and *Uintasorex*, *Sciuravus*, *Herpetotherium* and *Apatemys* are taxa characteristic of the Bridgerian. The co-occurrence of these taxa between the two areas, supports the temporal equivalence of the parts of the stratigraphic sequence in which these assemblages occur. Shoshonian (Earliest Uintan) mammals occur in Bone Beds A, B, and D, Unit 24 of Love (1939) in the type section of the Tepee Trail Formation, at other localities stratigraphically lower in the Tepee Trail Formation (e.g. Bare A. Locality), of the East Fork Basin area, and in the upper part of the Friars and lower part of the Mission Valley Formations in the San Diego area. All of these assemblages are from horizons that lie within an interval of reversed magnetic polarity. The upper lower Adobe Town Member assemblage is of equivocal age, but it shares the occurrence of several taxa with the Shoshonian (Earliest Uintan) assemblages from the San Diego and East Fork Basin areas.

The assemblages from the Aycross Formation, East Fork Basin area, and

from the Kinney Rim Member and the lower lower Adobe Town Member, Washakie Formation have been shown above to be Bridgerian in age. These two assemblages share the common occurrence of *Palaeosyops*, *Uintatherium*, *Patriofelis*, and *Hyrachyus*. The middle Adobe Town Member is clearly early Uintan (correlative with Uinta B strata) in age, and is younger than any assemblage described from the East Fork Basin and San Diego areas.

Therefore, the Bridgerian/Shoshonian (Earliest Uintan) boundary lies stratigraphically below the Bone Bed A and the Bare A. Locality, type section of the Tepee Trail Formation, East Fork Basin, Wyoming; below the mammal faunal horizons of the Friars and Mission Valley Formations, San Diego, California; and within the middle to upper part of the lower Adobe Town Member, Washakie Formation, Washakie Basin, Wyoming.

All the available data indicate that both the Bridgerian/Uintan boundary and earliest Uintan faunas occur within the same interval of reversed magnetic polarity in all three field areas.

Although there are significant faunal differences between the mammalian assemblages from the East Fork Basin area, Washakie Basin, and San Diego area there is sufficient information available to allow a relatively precise faunal correlation of sequences from these areas. The discussion above indicates that: (1) there is an interval of time, Shoshonian Land Mammal Subage (Earliest Uintan), represented in the strata from these areas that may be clearly defined and recognized faunally in the East Fork Basin and San Diego area sequences; (2) superposition of Bridgerian and Earliest Uintan or Early Uintan faunas in the East Fork Basin area and the Washakie Basin permit relatively precise placement of the Bridgerian/Uintan boundary in these sections; (3) either Shoshonian (Earliest Uintan) faunas and/or the Bridgerian/Uintan boundary can be recognized in the sequences from all three field areas; (4) in all three areas both earliest Uintan faunas *and* the Bridgerian/Uintan boundary occur *within* the same long interval of reversed magnetic polarity. What magnetic polarity chron does this long reversed polarity interval represent?

IDENTIFICATION OF MAGNETIC POLARITY INTERVALS

Unique identification of the magnetic polarity intervals from the three field areas of this study is not possible through use of magnetic polarity patterns alone. None of the sections studied provide a long enough sequence of polarity intervals to be matched unambiguously to the standard GMPTS based on unique correspondance of the lengths and pattern of polarity intervals. Therefore, an alternative technique must be used for this correlation.

The intertonguing, fossiliferous marine and continental strata of the San Diego area provide the critical information for the identification of the polarity intervals in the sequences from that field area. By mammalian biochronologic correlation of the San Diego area sections to the Wyoming sections, the identification of the San Diego polarity sequence may be used to identify uniquely the polarity intervals in the Wyoming sequences.

Berggren et al. (1985, figs. 3—6; tables 3 and 4) summarize all the available information that directly integrates and interfaces magnetic polarity sequencing and biostratigraphy (including about 200 Cenozoic planktonic datum events) for the Paleogene in European stratotype sections, deep sea Hydraulic Piston Cores (HPC's), and terrestrially-exposed marine sections. Magnetic polarity intervals are directly associated with marine biostratigraphic information in the San Diego area stratigraphic sequences. The now known correlation between marine biostratigraphy and magnetic polarity sequencing can be used to identify uniquely the magnetic polarity intervals in the San Diego sections.

Studies near Gubbio, Italy (Lowrie and Alvarez, 1981; Lowrie et al., 1982), in which the planktonic foraminiferal and calcareous nannoplankton zonations are directly associated with magnetic polarity stratigraphy sequences, locate the early/middle Eocene boundary at Chron C22N. The *Hantkenina aragonensis* (P10) Zone spans the time represented by the very youngest part of Chron C22N, the reversed interval Chron C21R, all of Chron C21N, and the oldest part of Chron C20R (Lowrie et al., 1982). The correlation of Zone NP14 with the magnetic polarity sequence is less precisely controlled, but Zone NP14 would have a maximum range of somewhere within Chron C22R to the base of the reversed interval of Chron C20R (Lowrie et al., 1982).

Evidence from DSDP Leg 73 (Hsu et al., 1982; Poore et al., 1983; Tauxe et al., 1983) supports these biostratigraphic/magnetostratigraphic associations. Calcareous nannoplankton zonations from Site 523 associate Zone NP15 with the reversed interval of Chron C20R, and part of Chron C20N. Much of the reversed interval of Chron C20R appears to be represented at Site 523 (although neither the top of Chron C21N nor the base of Zone NP15 is present because of missing section at the bottom of this site), and it is associated only with Zone NP15. Therefore, the top of Zone NP14 probably does not extend much above the base of the reversed interval Chron C20R.

In Berggren et al. (1985), fig. 5) the range of the *Hantkenina aragonensis* (P10) Zone extends from the very top of Chron C22N to the upper part (approximately the upper 1/4) of Chron C21N, and the range of the *Globigerinatheka subconglobata* (P11) Zone is from the upper part of Chron C21N to the lowest part of Chron C20N. Zone NP14 extends from the base of Chron C22N to the lower third of Chron C21N, and Zone NP15 extends from the lower third of Chron C21N to approximately the midpoint of Chron C20N.

Based on this information, the normal polarity interval (spanning the entire sampled thickness of the type section) of the type section of the Ardath Shale represents Chron C21N. The biostratigraphic information from the Ardath Shale indicates an age younger than the extreme base of Zone P10 for the entire normally magnetized thickness of the Ardath Shale type section (e.g. possible Zone P10/P11 age for the section; presence of Ulatisian/Narizian boundary — approximately equivalent to the Zone NP14/15

boundary — near the top of, or just above, the Ardath Shale). Since the base of Zone P10 only barely falls within the top of Chron C22N, and since the Zone NP14/15 boundary lies within Chron C21N, it is almost certain that the Ardath Shale normal polarity interval can *only* be correlated with Chron C21N.

The remainder of the San Diego area section probably also is middle Eocene in age (Lutetian to Bartonian, see above). The poorly defined "possible" P13 (middle Eocene) planktonic foraminiferal fauna from the Stadium Conglomerate, and the sparse nannofossil floras of late middle to "early late" Eocene age (probably equivalent to Bartonian = later *middle* Eocene in age) from the Stadium Conglomerate and Mission Valley Formations may indicate a middle Eocene Lutetian to possibly Bartonian age for these two formations. As was mentioned above, the Narizian/Refugian boundary is approximately equivalent to the middle/late Eocene boundary. There are no Refugian faunas in the San Diego area section, which supports an entirely middle Eocene (Lutetian to Bartonian) age for this sequence.

The Scripps and Friars Formations generally lie stratigraphically above the Ardath Shale and below the Stadium Conglomerate and Mission Valley Formation. By inference, the Scripps and Friars Formations are younger than Zone P10 (or possibly part of Zone P11) and Zone NP14 and possibly older than Zone P13 and late Lutetian or Bartonian. This inferred, broad age range possible for the Scripps and Friars Formations extends from the upper part of Chron C21N to the upper part of Chron C19N. Chron C20N extends from the base of Zone P12 to the middle of Zone P12, and from the upper part of Zone NP15 to about the middle of Zone NP16; it is correlative with later Lutetian middle Eocene. Identification of the Ardath Shale normal polarity interval as Chron C21N and the Friars/Mission Valley Formation normal polarity interval as possibly Chron C20N is consistent with the known and inferred age information for the San Diego area strata. Because of stratigraphic complexities, correlation of the sampled Friars Formation and Mission Valley Formation sections is quite difficult. Therefore, it is possible that *two* normal polarity intervals are present, one in the uppermost Friars and lowest Mission Valley and the other in the uppermost Mission Valley Formation. These two normal polarity intervals probably would be identified as Chrons C19N and C20N.

Based on these normal polarity interval identifications, it is clear that the reversed polarity interval present through most of the Delmar Formation is equivalent to Chron C21R, and the normal polarity interval at the top of the section is Chron C21N. The Delmar Formation (in the area sampled) is therefore entirely early middle Eocene (early Lutetian) in age. The normal polarity interval in the upper Scripps Formation and lower Friars Formation is Chron C21N correlative, and the reversed interval in the main body of the Friars Formation is Chron C20R.

The Shoshonian (Earliest Uintan) mammal faunas of the upper Friars and lower Mission Valley Formations first occur within the reversed interval

correlated with Chron C20R, and they possibly also occur within Chron C20N. Although the sparse fauna from the Scripps/Ardath Shale Formations have been reported as possibly Bridgerian in age, the Bridgerian/Uintan boundary is not definitely identifiable, faunally, in this sequence. However, because it is older than the Shoshonian (Earliest Uintan) fauna in the Friars Formation, the Bridgerian/Uintan boundary cannot be younger than Chron C20R.

In the East Fork Basin area a Shoshonian (Earliest Uintan) fauna, and the Bridgerian/Uintan boundary lie within a long reversed polarity interval (polarity zone E— of Fig.6). In both this area and the San Diego area Shoshonian (Earliest Uintan) faunas first occur within a long reversed polarity interval. Based on the faunal correlation (temporal equivalence of the Shoshonian, Earliest Uintan, faunas) between the two areas, polarity interval E— of the East Fork Basin area sequence must be correlative with Chron C20R. Similarly, the East Fork Basin area polarity interval D+, at the base of the Tepee Trail Formation and near the top of the Aycross Formation, must be Chron C21N. The correlation of the normal polarity interval, B+, lower in the Aycross Formation section is equivocal. The entire Aycross Formation normal polarity interval (B+ to D+) could represent Chron C21N with a very stratigraphically expanded, short temporal duration reversed event (C—) preserved between them (note the short reversed interval preserved in Chron C21N of the Contessa Highway section of Lowrie et al., 1982, fig.4); the lower normal (B+) could represent the preservation of an expanded, short normal event in Chron C21R; or the two normal polarity intervals, D+ and B+, could represent Chron C21N and Chron C22N, respectively. Although no definite resolution of this problem is presently possible, I believe the available paleomagnetic pattern data in this section and radioisotopic data on Bridgerian sediments argue against interpreting the lower normal polarity interval (B+) as Chron C22N. Interpretation of the Aycross Formation polarity sequence as representing part of Chron C20R (E—), all of Chron C21N (D+), all of Chron C21R (C—), all of Chron C22N (B+), and part of Chron C22R (A—) (an interval of at least 4.5 m.y. duration) is difficult to reconcile with the short temporal duration indicated by the radioisotopic dates on Bridgerian strata (see Table V).

In the Washakie Formation paleomagnetic section, classic Uintan faunas occur within a long normal polarity interval (C+), while classic Bridgerian faunas occur within the underlying, predominantly reversed polarity interval (B—). Undescribed faunas of possibly Earliest Uintan (Shoshonian) age occur within interval B_5—. Again, consistent with the data from the East Fork Basin and San Diego areas, the Bridgerian/Uintan boundary lies within a long reversed magnetozone (B_5—) (as does Earliest Uintan, by inference). This reversed polarity interval in the Washakie Formation, B_5—, is at least partly temporally correlative with the reversed polarity interval, E—, in the East Fork Basin area section and with the Friars Formation reversed polarity interval (C—) in the San Diego area section, based on faunal correlation. This

reversed interval is therefore correlative with Chron C20R. The overlying normal polarity interval in the Washakie Basin section, C+, is correlative with Chron C20N, and the highest reversed polarity interval, D−, is correlative with Chron C19R.

Correlation of the polarity intervals below C+ in the Washakie Formation is more difficult. The entire B− interval (B_1− to B_5−) may be equivalent to Chron C20R, the two short B+ intervals (B_2+, B_4+) may represent short duration normal polarity events within the reversed interval, and interval A+ may represent Chron C21N. Bridgerian faunas are present within much of the B− interval, and because Bridgerian is temporally short, I believe it is possible that all of B− is correlative with Chron C20R.

Alternatively, normal polarity interval B_2+ may be correlative with Chron C21N and polarity interval B_3− to B_5− alone would be equivalent to Chron C20R. The reversed polarity interval below B_2+ would either represent Chron C21R (and A+ would be equivalent to Chron C22N), or it may represent an expanded, short duration reversed polarity event within Chron C21N as was discussed above for the East Fork Basin area interval C− (and A+ would still be equivalent to part of Chron C21N). The presence of the "lower brown sandstones", and a basin-wide unconformity, at the top of B_2+ in the Washakie Basin section may support its identification as part of Chron C21N, with a significant portion of Chron C21N time missing from the section because of erosion and non-deposition. However, the presence of Bridgerian faunas in the section above and below B_2+, and the short temporal duration of Bridgerian time, argues against a significant temporal gap. A definite choice between these alternatives is not possible with the available data.

Shoshonian (Earliest Uintan) faunas and the Bridgerian/Uintan boundary consistently occur within a reversed polarity interval in the three Wyoming and California sequences. Marine biostratigraphic correlation of the San Diego sections to standard deep sea sequences (DSDP and Gubbio results) indicate that this reversed interval is correlative with Chron C20R, while the immediately underlying normal polarity interval in these sections is equivalent to Chron C21N.

Figure 9 presents a brief, graphic summary of the available radioisotopic, biostratigraphic and magnetostratigraphic data from the three field areas, and the proposed correlation between these areas.

CORRELATION OF MARINE AND CONTINENTAL CHRONOLOGIES

Wood et al. (1941) developed a provincial time scale for the North American continental Tertiary, based on "purely temporal" mammalian faunal units. That time scale was intended to be used only for provincial correlation, and precise correlation of the temporal units (Land Mammal Ages) to the European standard chronology was intentionally, and explicitly, avoided. The correlation chart of Wood et al. (1941, pl.1) did, regrettably,

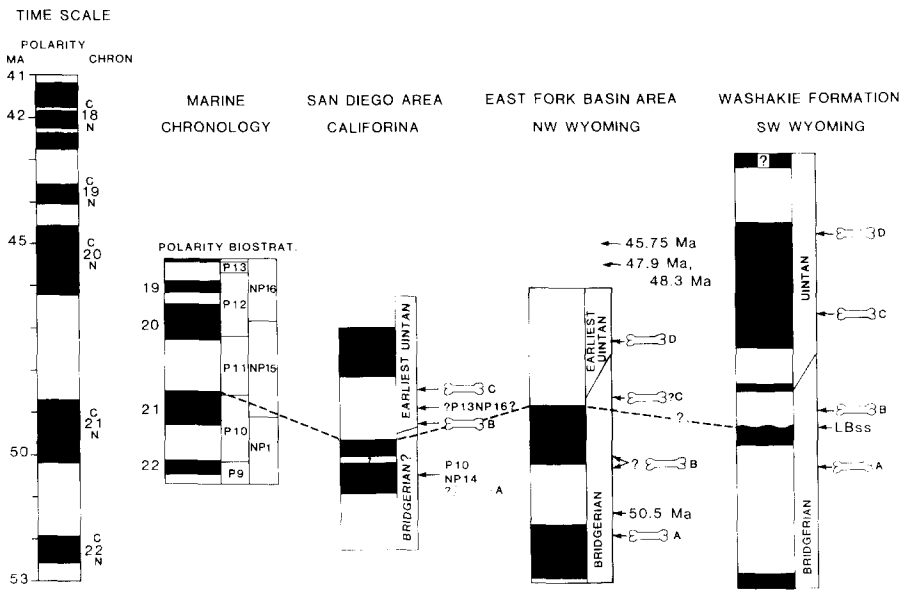


Fig.9. Composite correlation of the three field areas. The composite magnetic polarity sequences are plotted for each field area. To the right of each column is a graphic summary of biochronologic and radioisotopic data available from each sequence: Land Mammal Age data is given immediately to the right of each column (with inferred position of the Bridgerian/Uintan boundary), bones indicate position of mammalian faunas (lettered horizons discussed in text), radioisotopic dates are given in Ma, and *P* (planktonic foram) and *NP* (calcareous nannoplankton) zonal age assignments are presented for the San Diego area section. The marine chronology column is adapted from Berggren et al. (1985); it indicates the precise interrelationship between the magnetic polarity sequence and marine biochronologic zonations. These relationships are used to identify the polarity intervals in the San Diego area sequence, and by correlation, in the Wyoming sequences (see text). The time scale is from Berggren et al. (1985). The dotted line is the inferred correlation of the top of anomaly 21 (Chron C21N) in each section. Stratigraphic thicknesses and temporal intervals are drawn to different scales in each column; relative thicknesses and temporal durations of polarity intervals *within* each section are correct.

include an inferred correlation between the North American Provincial Ages and the standard European ages and epochs. Many subsequent biostratigraphers ignored the disavowal of equivalence (Wood et al., 1941) and began to invoke precise correlation between many of the European and North American ages. For instance, the Bridgerian was portrayed as middle Eocene and Uintan was considered late Eocene in the standard, global geochronology. Such uncritical acceptance of precise correlation between these ages was common, even though these correlations had never been documented or supported by precise and accurate data.

Based on biostratigraphic information from the San Diego area Golz (1976) and Golz and Lillegraven (1977) proposed that the Uintan in North America might actually be equivalent to some portion of the middle Eocene, rather than the late Eocene, of Europe.

Berggren et al. (1978) proposed a recalibration of Eocene North American Land Mammal "Ages" and the standard European Ages. Their evaluation of all the available marine and continental biostratigraphic information, and high temperature radioisotopic dates on biostratigraphic units and boundaries, allowed them to document precise correlations between the North American and European ages. In that study they equated all of the Bridgerian *and* all of the Uintan with the middle Eocene (Lutetian + Bartonian) of Europe. The Duchesnean (previously considered "late Eocene") was considered partly middle Eocene and partly late Eocene in age, and the Chadronian (previously considered entirely "early Oligocene" in age) was shown to be partly late Eocene and partly early Oligocene in age (Berggren et al., 1978, fig.2).

The data gathered in this study allows a further refinement of the correlation between North American mammalian chronology and the standard European marine chronology. The Bridgerian/Earliest Uintan boundary has been shown to lie within the reversed interval Chron C20R between Chrons C20N and C21N. This reversed polarity interval lies entirely within the temporal span of both Zone P11 and Zone NP15 in the marine biochronology. Therefore, the Bridgerian/Earliest Uintan boundary is clearly early Lutetian in age, and falls within biochronologic Zones P11 and NP15. Shoshonian (Earliest Uintan) time is definitely equivalent to part of the middle Eocene (as is most of the remaining portion of the Uintan by inference — see Berggren et al., 1978, 1985; and West et al., in press). The temporal equivalence for the entire Bridgerian is uncertain, but it is at least partially equivalent to the middle Eocene. Given the short temporal span of the Bridgerian (discussed below) it is likely the Bridgerian is entirely middle Eocene in age (as there are 4—4.5 m.y. between the Bridgerian/Uintan boundary within Chron C20R and the base of the middle Eocene at the very top of Chron C22N).

RADIOISOTOPIC AGE ESTIMATES FOR CHRON C21N AND THE BRIDGERIAN/UINTAN BOUNDARY

I have shown above that the Bridgerian/Uintan boundary and the younger boundary of Chron C21N occur within the stratigraphic sections sampled in this study. One goal of this project is to provide a radioisotopic age estimate for these boundaries. Several approaches may be used to determine such an age estimate; I consider two below.

Numerous high temperature radioisotopic dates have been determined for horizons within Bridgerian and Uintan strata of the western United States (see Table VI; and West et al., in press). Most of these radioisotopic dates are from sections that contain associated mammalian assemblages, but that do not have magnetostratigraphic information. Many of these dates are closely associated with, and provide good age approximations for, biochronologic units and boundaries. Almost all provide at least broad age constraint on the Bridgerian and Uintan.

There are at least eight high-temperature K—Ar dates on horizons of certain Bridgerian age from western Wyoming. These dates range from 48.9 to 50.5 Ma. Six other dates on samples from latest Wasatchian or early Bridgerian strata range from 49.1 to 50.6 Ma. Five dates from strata of Bridgerian or early Uintan strata range from 46.6 to 50.6 Ma, while eight dates from definitely Uintan or younger strata range from 44.7 to 48.3 Ma. The large number of dates within and bracketing the Bridgerian indicate that the Bridgerian was probably quite short temporally, lasting only 2—3 m.y. (as noted by McKenna et al., 1973 and Berggren et al., 1978), from about 51—52 Ma to about 48—49 Ma. Numerous dates tightly constrain the age of the Bridgerian/Uintan boundary to between 48 and 49 Ma (estimates based on recalculated dates). The available age data from strata that are clearly Bridgerian or clearly Uintan indicate an age of approximately 48.5—49 Ma for the Bridgerian/Uintan boundary. West et al. (in press) estimate the age of the Bridgerian/Uintan boundary at about 48.5 Ma.

The younger boundary of Chron C21N lies within the Bridgerian, and an age estimate for the top of Chron C21N must be older than the age estimates cited above for the Bridgerian/Uintan boundary. Based on my age estimate of about 49 Ma (or 48.5 Ma based on the conclusions of West et al., in press; both derived from corrected dates) for the Bridgerian/Uintan boundary, and the radioisotopic date range of 49—50.5 Ma for Bridgerian age strata, the younger boundary of Chron C21N could be inferred to be about 49.5—50 Ma.

Direct association of radioisotopic dates, magnetostratigraphy, and biostratigraphy in the East Fork Basin area permits a more precise estimate of the age of the younger boundary of Chron C21N, and the Bridgerian/Uintan boundary. In the East Fork Basin area, three horizons bracketing the top of Chron C21N have provided four high-temperature K—Ar ages (Smedes and Prostka, 1972; Love et al., 1978). The dated horizons are within the same stratigraphic sections that were magnetostratigraphically sampled and that contain Bridgerian and Uintan faunas and the Bridgerian/Uintan boundary. This direct association of radioisotopic dates, magnetostratigraphy and biostratigraphy permits a precise estimate of the ages of the magnetic polarity and biochronologic boundaries present in this section.

The stratigraphically lowest horizon lies within the Aycross Formation paleomagnetic section, approximately 950 ft (289.5 m) below the top of the normal polarity interval correlated with Chron C21N. The K—Ar age on biotite from this horizon is 50.5 ± 0.5 Ma. Two dates of 47.9 ± 1.5 Ma and 48.3 ± 1.3 Ma (mean = 48.1 Ma) have been determined on biotites within the Wiggins Formation, from a single horizon 500—600 ft (152—183 m) above the local top of the Tepee Trail Formation in the Tepee Creek section. The fourth date of 45.75 ± 1.2 Ma is on a biotite from a sample 650—750 ft (198—213 m) above the local top of the Tepee Trail Formation in the same section. The two Wiggins Formation horizons are from strata less than 1 km away from the Tepee Creek section, and can be easily tied in stratigraphically at approximately 1650 ft (503 m) and 1800 ft (549 m) above the top of the

TABLE VI

High temperature, K—Ar dates from Bridgerian/Uintan strata, Wyoming. All dates in Ma. "Recalibrated date" refers to recalculation of "original date" (as cited in the reference list, last column) using the revised K—Ar decay and abundance constants of Steiger and Jager, 1977. The best available determination of the biochronologic age of the dated strata is given in the "temporal significance" column

Recalibrated date (Ma)	Original date (Ma)	Lithologic unit	Other information	Temporal significance	Reference
50.3	49.0	Unit 1, Wagonbed Fm.	KA 1012	probable early Bridgerian	Evernden et al., 1964
50.5	49.2	Halfway Draw Tuff, Wind River Fm.	KA 1021	Late Wasatchian	Evernden et al., 1964
46.6	45.4	Unit 3, Wagonbed Fm.	KA 1018	(?) Late Bridgerian/ early Uintan	Evernden et al., 1964
46.2	45.0	"Tepee Trail Fm.", Badwater	KA 1024	(?) Early Uintan	Evernden et al., 1964
47.4 ± 1.8	46.2 ± 1.8	Base of Wiggins Fm.	Lava flow, Pinnacle Buttes area	Bridgerian, or younger	Rohrer and Obradovich, 1969
47.7 ± 2.3	46.5 ± 2.3	Base of Wiggins Fm.	Boulder in conglomerate overlying lava (5′ above), Pinnacle Buttes area	Bridgerian, or younger	Rohrer and Obradovich, 1969
48.9 ± 1.5	47.7 ± 1.5	7 m above Bulldog Hollow Mbr., Fowkes Fm.	State Line Quarry	Bridgerian	Oriel and Tracey, 1970
50.5 ± 1.5	49.2 ± 1.5	Lost Creek Tuff Mbr., Sepulcher Fm.	Overlies Willwood Fm.	Latest Wasatchian, or younger	Smedes and Prostka, 1972
49.2 ± 1.3	48.0 ± 1.3	Pacific Creek Tuff Mbr., Trout Peak Trachyandesite	Overlies Willwood Fm.	(?) Bridgerian	Smedes and Prostka, 1972
49.1 ± 1.3	47.9 ± 1.3	"Two Ocean Fm."	Overlies Wasatchian or Bridgerian, Underlies correlated Tepee Trail Fm. (Bridgerian or Uintan)	probable Bridgerian	Smedes and Prostka, 1972
49.0 ± 1.3	47.8 ± 1.3	Tepee Trail Fm. (?)	1/2 mile NW of Togwotee Pass, 230′ below Bridgerian fauna	probable Bridgerian	Smedes and Prostka, 1972
49.1 ± 1.3	47.9 ± 1.3	Tepee Trail Fm. (?)			
50.6 ± 1.4	49.3 ± 1.4	"Tepee Trail Fm."	Bentonitic ash, 100′ below top of fm., Pinnacle Buttes area	Bridgerian or early Uintan	Smedes and Prostka, 1972
47.3 ± 1.2	46.1 ± 1.2	"Tepee Trail Fm."	Ash 10′ below top of fm., Pinnacle Buttes area	Bridgerian or early Uintan	Smedes and Prostka, 1972

45.6 ± 1.4	44.4 ± 1.4	Wiggins Fm.	Air-fall tuff, 800′ above base of fm., Pinnacle Buttes area	(?) Uintan, or younger	Smedes and Prostka, 1972
46.7 ± 1.3	45.5 ± 1.3	Wiggins Fm.	Air-fall tuff, 800′ above base of fm., Pinnacle Buttes area	(?) Uintan, or younger	Smedes and Prostka, 1972
48.3 ± 1.3	47.1 ± 1.3	Wiggins Fm.	500—600′ above base of fm., Wiggins Fork area	Early Uintan, or younger	Smedes and Prostka, 1972
47.9 ± 1.5	46.7 ± 1.5	Wiggins Fm.	500—600′ above base of fm., Wiggins Fork area	Early Uintan, or younger	Smedes and Prostka, 1972
45.8 ± 1.2	44.6 ± 1.2	Wiggins Fm.	650—750′ above base of fm., Wiggins Fork area	Early Uintan, or younger	Smedes and Prostka, 1972
50.6	49.3	White Pass bentonite, ?Aycross Fm.	Predates Kisinger Lakes flora, correlates with Loc. L-41 (McKenna, 1980)	Latest Wasatchian/ earliest Bridgerian	MacGinitie et al., 1974
	"about 50"	Lapilli tuff, correlated Aycross Fm.		Latest Wasatchian	MacGinitie et al., 1974
50.3	49	Laney Shale Mbr., Green River Fm., Green River Basin	Dates Little Mountain flora		MacGinitie et al., 1974
45.1 ± 2.6	44.0 ± 2.6	Rhyodacite from vents and intrusives cutting Wagon Bed Fm.	Rattlesnake Hills area	Uintan, or younger	Pekarek et al., 1974
44.7 ± 1.0	43.6 ± 1.0	Phonolite from vents and intrusives cutting Wagon Bed Fm.	Rattlesnake Hills area	Uintan, or younger	Pekarek et al., 1974
50.3 ± 0.5	49.2 ± 0.5	Type area of Aycross Fm.	70.7 m above flora near Bridgerian mammals	Bridgerian	Love et al., 1978
49.75 ± 0.5	48.5 ± 0.5	Pumice from tuff, Bridger Fm., Tabernacle Butte	Overlies late Bridgerian mammals	Bridgerian or younger	Berggren et al., 1978
49.1 ± 1.9	47.9 ± 1.9	7 m above Bulldog Hollow Mbr., Fowkes Fm.	State Line Quarry, Bridgerian mammals	Bridgerian	Nelson, 1979
47.9 ± 0.5		"Blue Point Cgl. Mbr.", Wiggins Fm., Carter Mtn.	Overlies Bridgerian mammals and Trout Peak trachyandesite	Bridgerian, or younger	Bown, 1982
48.5 ± 0.6		"Blue Point Cgl. Mbr.", Wiggins Fm., Carter Mtn.	Overlies Bridgerian mammals and Trout Peak trachyandesite	Bridgerian, or younger	Bown, 1982
49.2 ± 0.7		Wiggins Fm., near Sylvan Pass		?	Bown, 1982

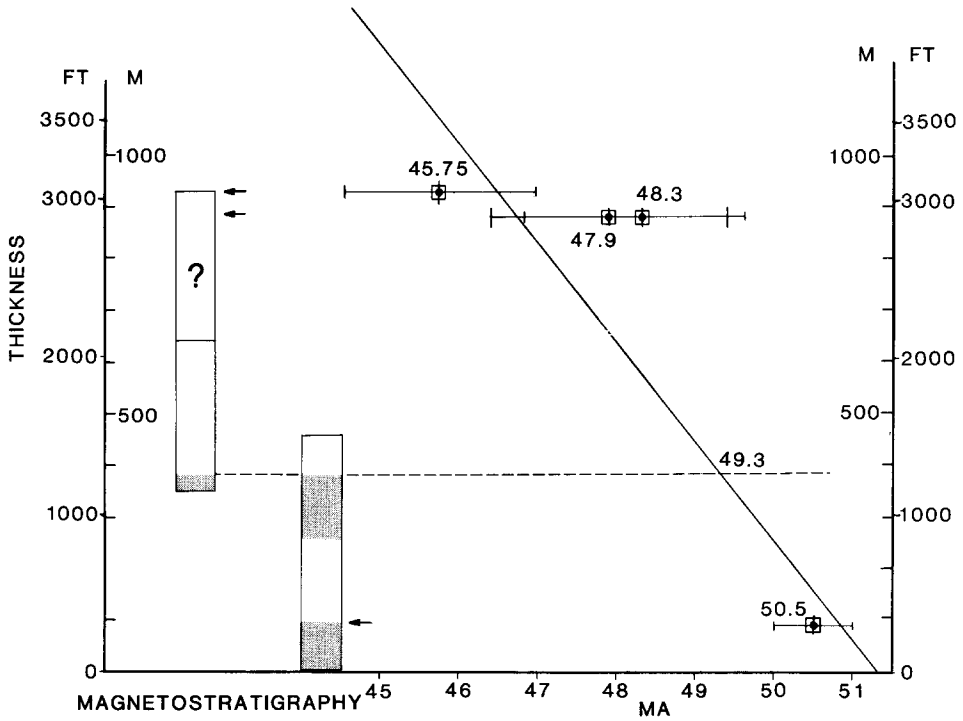


Fig.10. Radioisotopic age estimate of Chron C21N, East Fork Basin area, Wyoming. Left side of diagram indicates the composite magnetic polarity sequence plotted against stratigraphic thickness. Arrows to the right of the polarity columns indicate the stratigraphic position of the high temperature, K—Ar dates from this section. The question mark indicates that this stratigraphic interval (Wiggins Formation) has not been sampled paleomagnetically. Right side of diagram is a plot of age (in Ma) versus stratigraphic thickness. The four available radioisotopic dates (plotted as open squares, with a vertical line to indicate the date) are given with associated error bars. A least squares regression line through these dates (see text) gives an age estimate of 49.3 Ma for the top of Chron C21N.

Chron C21N boundary in the Tepee Creek/Aycross Formation paleomagnetic sections. Figure 10 summarizes the stratigraphic positions of the dated horizons and the magnetostratigraphic data from the paleomagnetic sections.

A best fit (least squares) regression line through all four of these dates (see McDougall et al., 1977 for description and discussion of this approach to generating stratigraphic age estimates) yields an age estimate for the younger boundary of Chron C21N (1265 ft, or 385.6 m, above the base of the section) of 49.3 Ma. If the $+X$ axis is in Ma and the $+Y$ axis is in distance above the base of the section, the formula for the best fit regression line would be:

Y (in feet above base) = $32095 - 625X$ (in Ma), or Y (in meters above base) = $9782.6 - 190.5X$ (in Ma)

As the age of 49.3 Ma for the younger boundary of Chron C21N is an age *estimate* derived from a statistical regression analysis, it is impossible to

assign a simple analytical error bar as if this age estimate were a directly experimentally derived radioisotopic date. I favor an age estimate of approximately 49.3—49.5 Ma for this boundary. It is rarely possible to achieve such precision in an age estimate for a Paleogene magnetic polarity interval boundary, because radioisotopic dates are generally not directly associated with precisely identified magnetic polarity sequences.

TIME SCALES AND GEOCHRONOLOGY

In the preceding sections I have developed, and detailed a correlation network that *directly* interfaces magnetostratigraphy, biostratigraphy (marine and continental), and radioisotopic chronology (high-temperature K–Ar dates). This correlation network provides results that have great significance for Cenozoic geochronology. No previous early Cenozoic study has been able directly to use high temperature radioisotopic ages for dating any associated biostratigraphic and polarity chron boundary. As such, all previous geochronologies have relied on either low temperature radioisotopic age determinations (mainly on glauconites, and *not* in sections directly associated with magnetic polarity data) or lengthy extrapolation/interpolation between well constrained calibration points (in a GMPTS), to determine the precise ages of biostratigraphic and magnetic polarity boundaries in the early Cenozoic. The data from the western United States, that I have discussed above, *do* allow relatively precise dating of the Bridgerian/Uintan Land Mammal Age boundary and the age of the younger boundary of Chron C21N. The best age estimate for the Bridgerian/Uintan boundary is approximately 49 Ma, while that for the younger boundary of Chron C21N is approximately 49.5 Ma. The potential accuracy of these age estimates is strongly supported by the fact that I have documented a single stratigraphic sequence in northwestern Wyoming that directly integrates magnetostratigraphy (including the younger boundary of Chron C21N), continental biostratigraphy (spanning the Bridgerian/Uintan boundary), and radioisotopic chronology (four independent high-temperature K—Ar dates bracketing these critical boundaries).

The age estimate of 49.5 Ma for the younger boundary of Chron C21N generated in this study is older than previous age estimates for this boundary by approximately 0.5 m.y. (LaBrecque et al., 1977) to 6 m.y. (Tarling and Mitchell, 1976). The results of this study support the accuracy of the LaBrecque et al. (1977) magnetic polarity time scale in this interval of early Cenozoic time — a time for which that time scale had no direct radioisotopic calibration. There is a significant difference between my age estimate of 49.5 Ma for the top of Chron C21N and the ages of 47.69 of Ness et al. (1980) and 47.37 Ma of Lowrie and Alvarez (1981). My results are extremely discordant with the ages of 45.7 Ma of Butler and Coney (1981) and 43.37 Ma of Tarling and Mitchell (1976) for this boundary. Figure 11 provides a comparison of these magnetic polarity timescales for part of the early Cenozoic (43—60 Ma). The results of this study may be used simply as

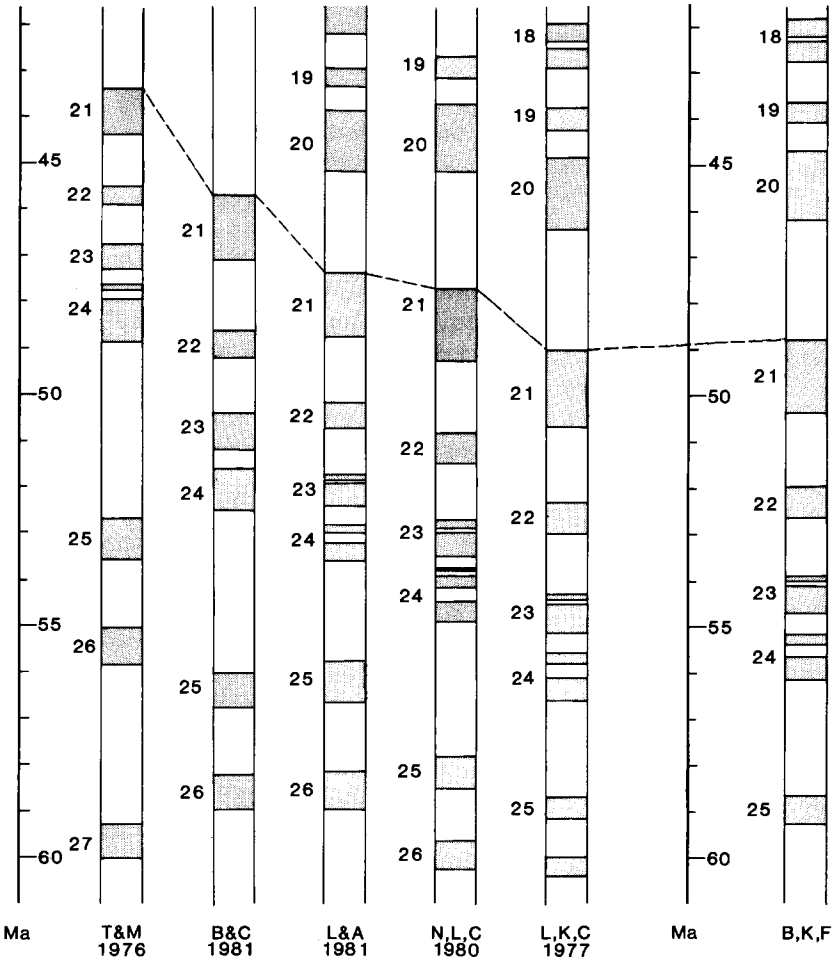


Fig.11. Comparison of magnetochronologies. Six magnetochronologies are presented for comparison of the ages assigned to Paleocene—Eocene magnetic anomaly (polarity chron) boundaries. The magnetochronologies are contrasted in the text. The age scales are in Ma, and the dashed line is a datum connecting the top of anomaly 21 (Chron C21N) in each scale. Anomaly (polarity chron) boundary ages were taken from tables within the texts of the following magnetochronologic and geochronologic studies: *T & M, 1976* = Tarling and Mitchell (1976); *B & C, 1981* = Butler and Coney (1981); *L & A, 1981* = Lowrie and Alvarez (1981); *N, L, C, 1980* = Ness et al. (1980); *L, K, C, 1977* = LaBrecque et al. (1977); *B, K, F* = Berggren et al. (1985).

a corroboration/contradiction test of the accuracy of previously generated time scales. In this case, the time scale of LaBrecque et al. (1977) is most strongly corroborated and the Tarling and Mitchell (1976) time scale is most strongly contradicted in the region of the younger boundary of Chron C21N.

The successful integration of many geochronological data sources in this

study, however, permits the use of the well-constrained age estimate on the younger boundary of Chron C21N as a calibration point in a GMPTS. Previous time scales have lacked calibration points in the early Cenozoic that directly integrate magnetic, biostratigraphic *and* radioisotopic information. Berggren et al. (1982, 1985; and see below) incorporate the age estimate for Chron C21N, generated in this study, as a critical calibration point in their recent geochronology. This geochronology closely resembles that of LaBrecque et al. (1977) for much of the middle Cenozoic, but differs significantly from many of the geochronologies cited above over the same interval (see Fig.11). Many of the discrepancies and similarities are due to variations in assumptions, methodologies and data sources among the different geochronologies (Flynn, 1983).

Berggren et al. (1982, 1985) propose a broad and synthetic geochronology that integrates extensive new data from marine and continental biostratigraphy, magnetic polarity sequencing, and radioisotopic chronology. We have rigorously evaluated the available radioisotopic age data that provide age estimates for biostratigraphic stage boundaries; these age estimates agree very closely with our magnetochronologic predicted ages for the stage boundaries (based on the known relationship between the biostratigraphic zonation and magnetic polarity sequence; see fig. 2, Berggren et al., 1985). These stage boundary age estimates alternatively could have been used as calibration points, and the resultant geochronology would have been very similar to the GMPTS we generated independent of these points. The close agreement between these independently derived age estimates and the predicted ages for the stage boundaries supports the validity of our geochronology.

As was mentioned above, much of the Cenozoic chronology of Berggren et al. (1985) yields a chronology that is very similar to that of LaBrecque et al. (1977), which used different calibration points and a somewhat different methodology. Several of the calibration points used by Berggren et al. (1985) to constrain the Cenozoic were only recently derived in this study and in Prothero et al. (1982), both of which integrate magnetochronology, biochronology and high-temperature radioisotopic chronology in terrestrial (and marine) sequences. The recent, very precise correlations between biostratigraphy and magnetostratigraphy demand that any chronology be based on a set of radioisotopic ages which is consistent with available age data from *both* systems; these chronologies must be integrated and mutually consistent. The recent radioisotopic chronologies of Odin and co-workers (e.g. Odin, 1975, 1978, 1982; Odin et al., 1978; used by Tarling and Mitchell, 1976 as calibration for their GMPTS; but see Berggren et al., 1985, appendix II) frequently yield internally consistent results, but these chronologies conflict strongly with several others (e.g. Berggren et al., 1985) that integrate radioisotopic data directly associated with both biostratigraphy and magnetostratigraphy.

It is important to note that although the radioisotopic age estimate for the

younger boundary of Chron C21N generated in this study is 49.5 Ma, the age of this boundary is 48.75 Ma in the Berggren et al. (1985) magnetobiochronology. This slight difference results from the use of this age estimate as only one data point constraining a least squares regression line segment drawn using several data points. This is a significant methodological difference from the approach used in other GMPTS studies. Those studies use age estimates on boundaries as calibration points *through* which their magnetochronologic correlation line (age versus composite magnetic polarity sequence) must pass. The Berggren et al. (1985) time scale used these age estimates as a broader control on defining linear correlation segments that are constrained by, but that are not *required* to pass through, several "calibration points". We do not assume that these points are precisely accurate, but that there is some inherent error (in the dating technique, interpolation/extrapolation techniques for estimating the age of a boundary stratigraphically separated from the dated horizon, etc.) associated with these age estimates.

Several other recent studies provide independent, high-temperature constraints on numerical ages of Paleogene chronologic intervals (Fig.12), Radioisotopic dates and magnetic polarity stratigraphies are available from early Paleocene strata and the Cretaceous/Tertiary boundary in Montana and Colorado, U.S.A. and Alberta, Canada (Lerbekmo et al., 1979; Baadsgaard and Lerbekmo, 1980, 1982; Archibald et al., 1982; Obradovich, 1984). In Montana and Alberta the Cretaceous/Tertiary boundary is marked by biological extinctions and turnovers (dinosaurs, mammals, plants), and an iridium anomaly. Obradovich (1984) provided an ^{40}Ar—^{39}Ar date of 66.0 ± 0.54 Ma on sanidine from the "Z-coal" in Montana. Although the Z-coal commonly is considered to mark the Cretaceous/Tertiary boundary, multiple "Z-coals" are known in Montana (Archibald et al., 1982). This date is from a horizon stratigraphically above another "Z-coal" that contains an iridium anomaly; the dated horizon probably lies at the base of a normal polarity interval correlative with the early Paleocene Chron C29N (J.D. Obradovich, personal communication, 1984). Based on his ^{40}Ar—^{39}Ar studies, Obradovich (1984) discussed likely explanations for the anomalously young dates of Baadsgaard and Lerbekmo (1980, 1982) on this same "Z-coal" and other early Paleocene horizons in Alberta. Three concordant dates of 66.3 Ma (K—Ar, plagioclase; Evernden et al., 1964), 65.8 ± 0.7 Ma (K—Ar, biotite; Obradovich and Cobban, 1975) and 65.8 ± 0.34 Ma (^{40}Ar—^{39}Ar, biotite; Obradovich, 1984) are available from early Puercan age strata, Colorado. The Puercan extends from the Cretaceous/Tertiary boundary (Chron C29R) to Chron C28N or C28R (see Berggren et al., 1985; Flynn et al., 1984). These early Paleocene dates (Fig.12, point *1*) tightly constrain the age of Chron C29N to be approximately 66 Ma and the Cretaceous/Tertiary boundary to be no less than 66.3 Ma. The predicted ages for these intervals in the Berggren et al. (1985) chronology are 65.5–66.2 Ma (Chron C29N) and 66.4 Ma (K/T boundary.

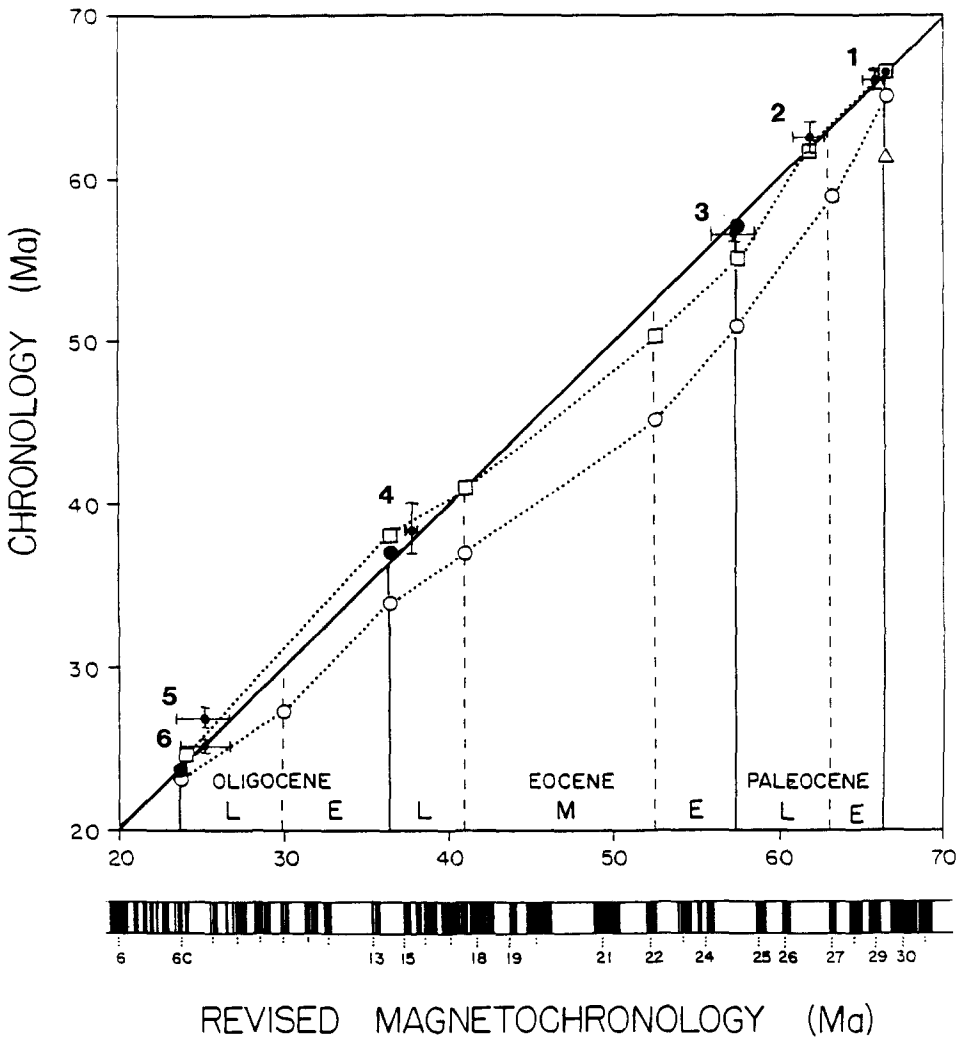


Fig.12. Comparison of available high temperature mineral dates associated with magnetochronologic and biochronologic intervals or boundaries versus the predicted ages in the Berggren et al. (1985) geochronology. If dates coincide with the predicted ages they should fall along the diagonal black line. Points *1*—*6*: dates referred to in text. Vertical error bars are analytical standard errors; horizontal error bars represent maximum uncertainty in biochronologic or magnetochronologic placement of these dates. Solid circles: Berggren et al. (1985), independent age data constraining ages of Epoch boundaries. Open circles: from Odin and Curry (1981). Squares: from Lowrie and Alvarez (1981). Anomaly numbers indicated below portrayal of geomagnetic polarity reversal sequence.

Marshall (1982) discussed a sequence of middle Paleocene, intertonguing marine and continental strata from South America. In that sequence two basalts in the basal Salamanca Formation gave whole rock ages of 64.2 ± 0.8 Ma (reported as 64.0 ± 0.8 Ma in Odin, 1982) and 62.3 ± 0.4 Ma. A

vitric tuff from the top of the Salamanca Formation (or the base of the conformably overlying Rio Chico Formation) has been dated as 62.5 ± 5 Ma. The Salamanca Formation contains planktic forams indicative of a late Danian age. A magnetic polarity sequence in the Rio Chico Formation has been correlated with Chron C26R (at the base of the formation) to Chron C25N (Marshall, 1982). The available dates provide a maximum age estimate of 62—63 Ma for Chron C26R. Berggren et al. (1985) present an age of 60.8—63.0 Ma for Chron C26R (Fig.12, point *2*).

In Greenland a sequence of reversely magnetized basalts has been correlated with Chron C24R (Beckinsale et al., 1970; Soper et al., 1976; Nielsen et al., 1981). A basalt near the top of the sequence has been dated at 56.5 ± 0.6 Ma (Fig. 12, point *3*; see discussion in Berggren et al., 1985). Marine interbeds bracketing this dated basalt contain dinoflagellate assemblages of late Paleocene age (below) and early Eocene age (above). This basalt date is consistent with the Berggren et al. (1985) age of 57.8 Ma for the Paleocene/Eocene boundary (mid- Chron C24R).

Prothero and Armentrout (1985) present a magnetic polarity stratigraphy of late Eocene—Oligocene marine strata from the Olympic Peninsula, Washington, U.S.A. These strata contain benthic foram assemblages of the Ulatisian to Saucesian Stages, and molluscan faunas of Galvinian to Pillarian Stages; the magnetostratigraphy for this sequence is correlated to Chrons C15R to C6CN. Correlation of the base of the sequence to Chron C15 is supported by calcareous nannoplankton assemblages assigned to zones NP19/20 and NP21. A high-temperature, K—Ar date of 38.5 ± 1.6 Ma has been determined on a basalt underlying basal Refugian Stage and Galvinian Stage strata (Fig.12, point *4*). This basalt falls within (or is possibly slightly older than) Chron C15R. This date is consistent with the Berggren et al. (1985) age of 37.7—38.1 Ma for Chron C15R.

High-temperature, K—Ar dates are available for late Oligocene strata from South America and Mexico. Marshall et al. (1977) considered the Monte Leon Formation, South America to be late Chattian in age based on molluscan and planktonic foram assemblages. The Monte Leon Formation is conformably underlain by the Colhue-Huapi beds which contain basalts with whole rock dates of 24.3 ± 0.5 Ma, 27.7 ± 0.6 Ma, and 28.8 ± 0.9 Ma (Fig.12, point *5* — mean of these three dates). A tuff at the base of the Santa Cruz Formation (conformably overlying the Monte Leon Formation) has been dated at 21.7 ± 0.3 Ma.

The San Gregorio Formation, Baja California Sur, Mexico contains latest Oligocene diatom assemblages and late Oligocene coccolith assemblages (McLean and Hausback, 1982). Interbedded tuffs provide four dates that range from 23.0 ± 0.5 Ma to 27.2 ± 0.6 Ma. The most reliable analysis appears to be the date of 25.3 ± 0.3 Ma on a high potassium (8.14% K_2O) biotite from a crystal vitric tuff (Fig.12, point *6*).

Although these late Oligocene dates are from strata that are not precisely constrained biostratigraphically or magnetostratigraphically, ages of 23—28 Ma

for late Chattian (latest Oligocene) strata are consistent with the Berggren et al. (1985) age of 23.7 Ma for the younger boundary of the Oligocene.

These high-temperature radioisotopic dates, from sequences that also contain biostratigraphic and magnetostratigraphic information, provide an independent test of the validity of the Berggren et al. (1985) magnetobiochronology. Berggren et al. (1985, fig. 2 and text) evaluated similar independent information, consisting primarily of radioisotopic dates associated with major biochronologic boundaries. The high temperature dates evaluated by Berggren et al. (1985) and in this study conflict with the consistently younger ages proposed in chronologies based primarily on low temperature, glaucony dates (e.g. Odin, 1982; Odin and Curry, 1981). The consistent agreement between these independent dates and the ages predicted by the Berggren et al. (1985) geochronology supports the validity of the methodology and the accuracy of the calibration-points (including the age of the younger boundary of Chron C21N determined in this study) used to construct that chronology.

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